UNIVERSITY OF ILLINOIS @ URBANA-CHAMPAIGN

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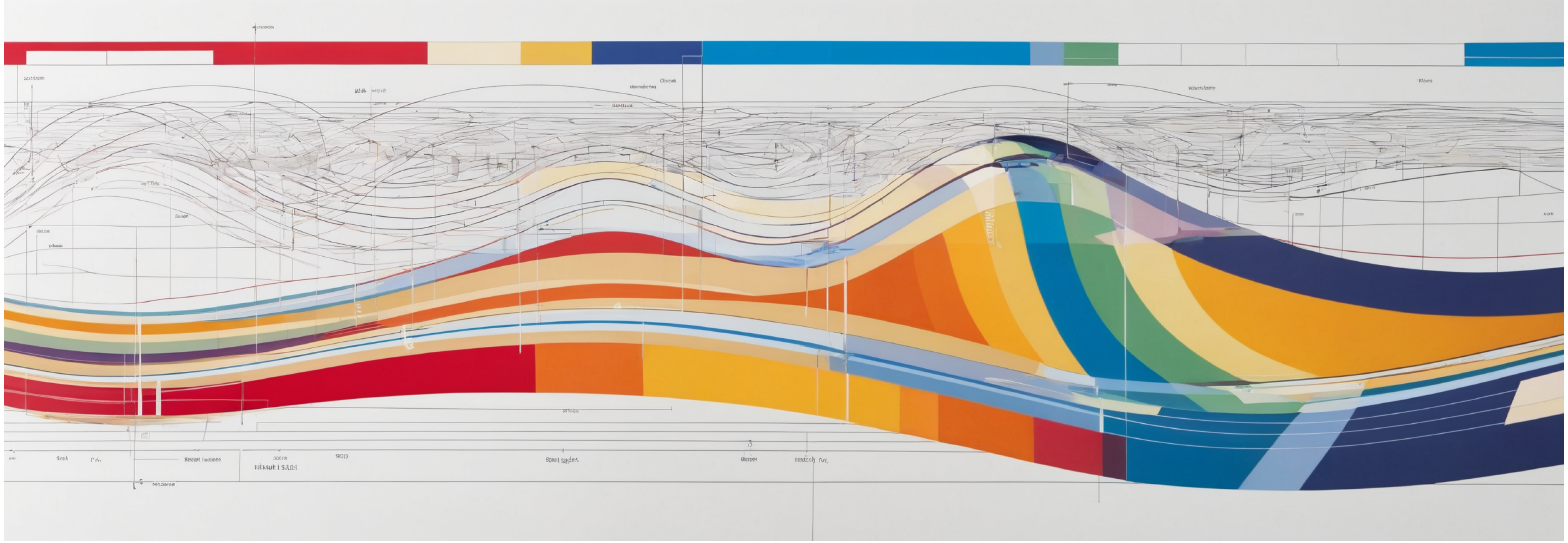

Audio Computing Lab

Pitch Tracking & Modifications

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Overview

- Pitch
  - Definition
  - Pitch tracking
    - Monophonic & polyphonic
- Pitch modifications
  - Pitch shifting
  - Auto-tuning

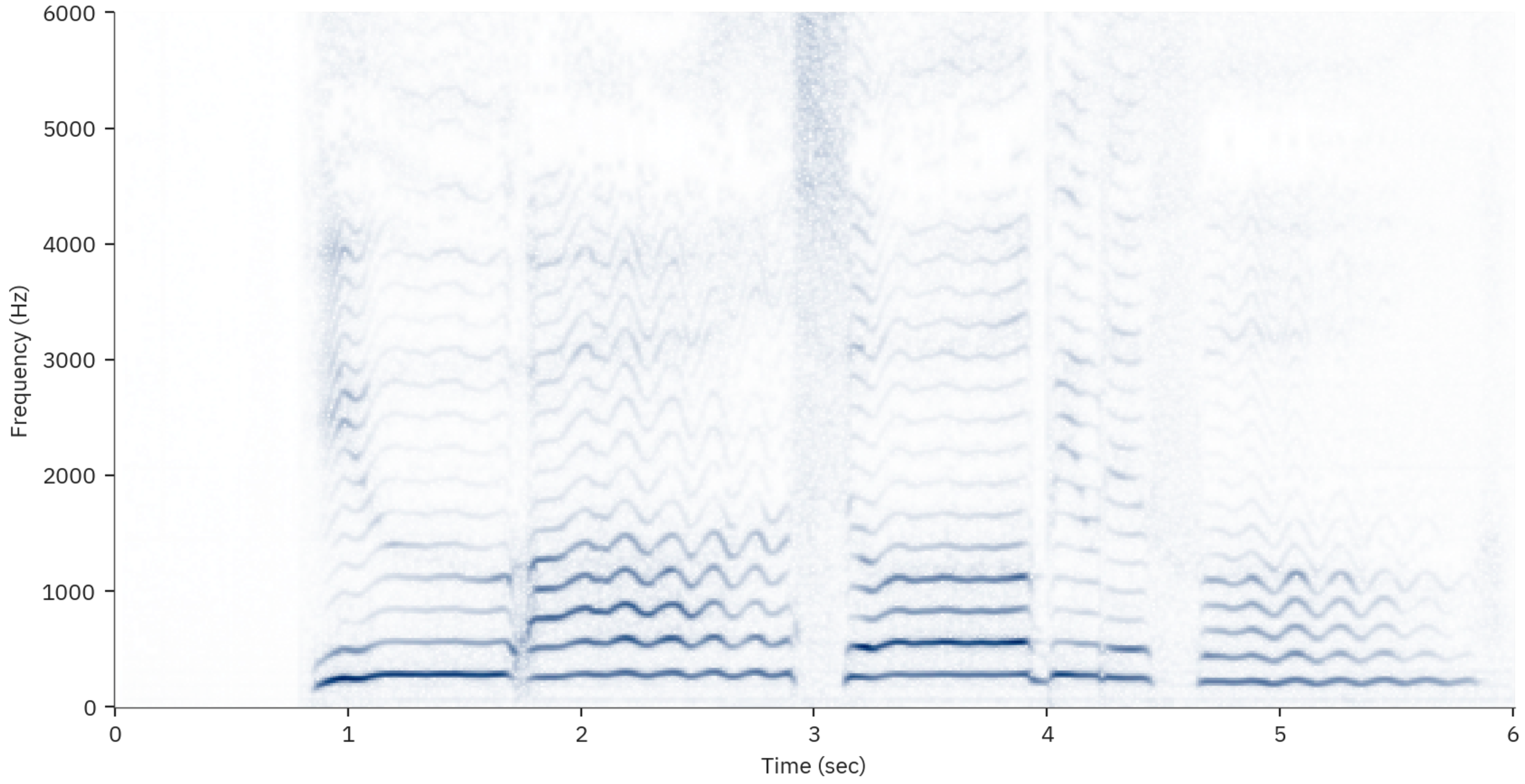
Some motivation

What is pitch?

- Depends on who you ask!
- Pitch is a *percept*, not a physical quantity
- Generally it relates to the rate of waveform repetition

Different names, same thing

- For speech we use the term *f-zero* (notated: $F0$, F_0, or f_0)
  - f_0 is the frequency of the lowest periodic element
    - f_1 is the first harmonic, etc …

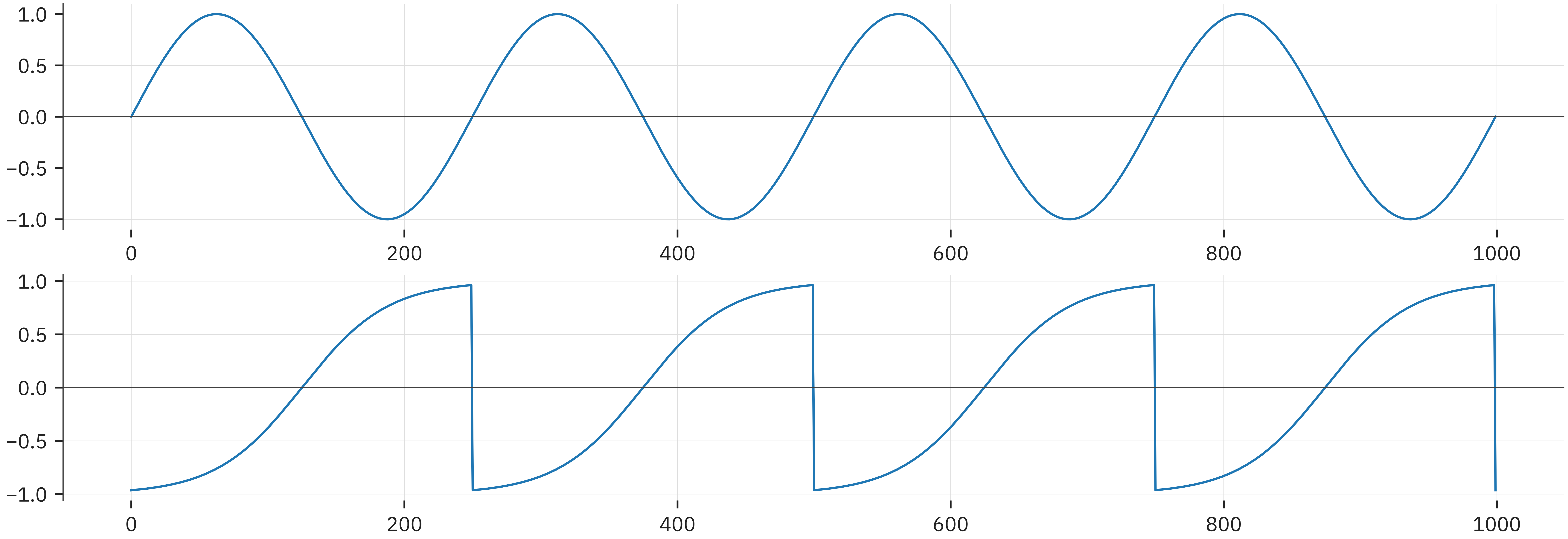


- For music we use notes
  - E.g. C♯, B♭, middle-C, F_3, etc.
  - Or MIDI note values: 0-127

- We also use the term *fundamental frequency*

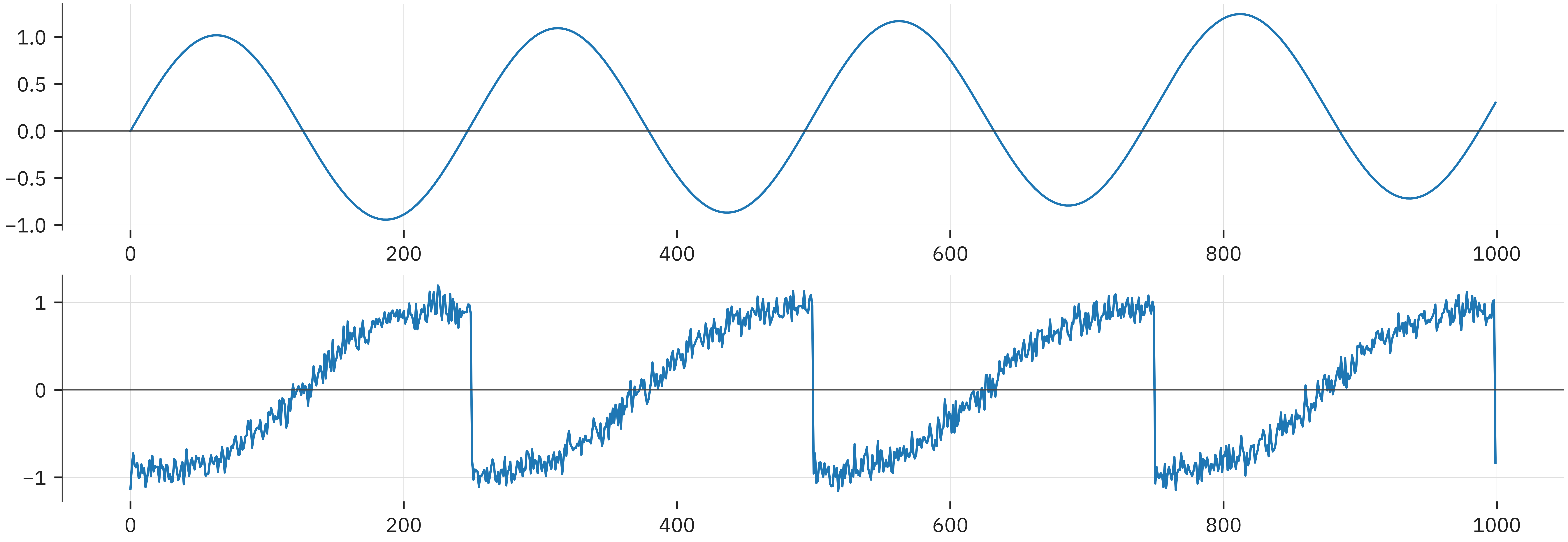
A formal definition

- *Periodicity:* $s[t+P] = s[t]$
  - Function repeats every P samples, P defines pitch
    - Is that realistic?



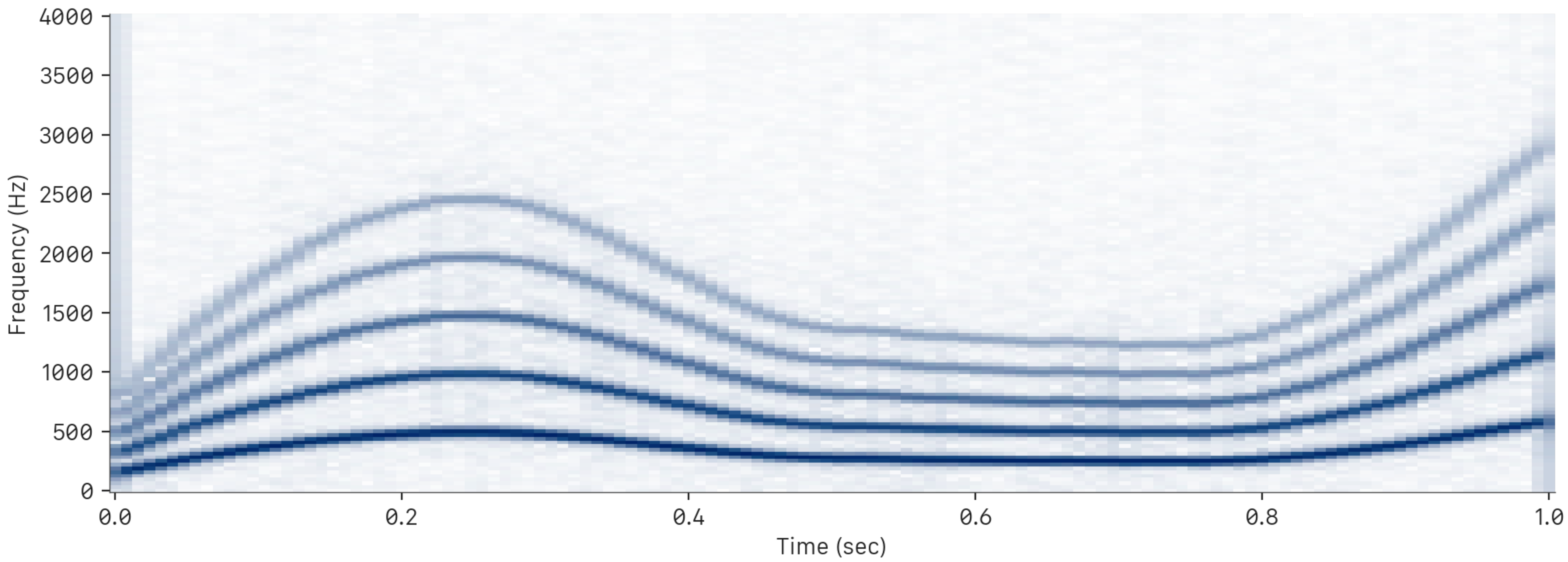
A more real-world version

- *Quasi-periodicity:* $s[t+P] = f(\, s[t] \,)$
  - Where $f(x) = x+C$, or $f(x) = ax$, …
    - i.e. almost periodic, but not quite

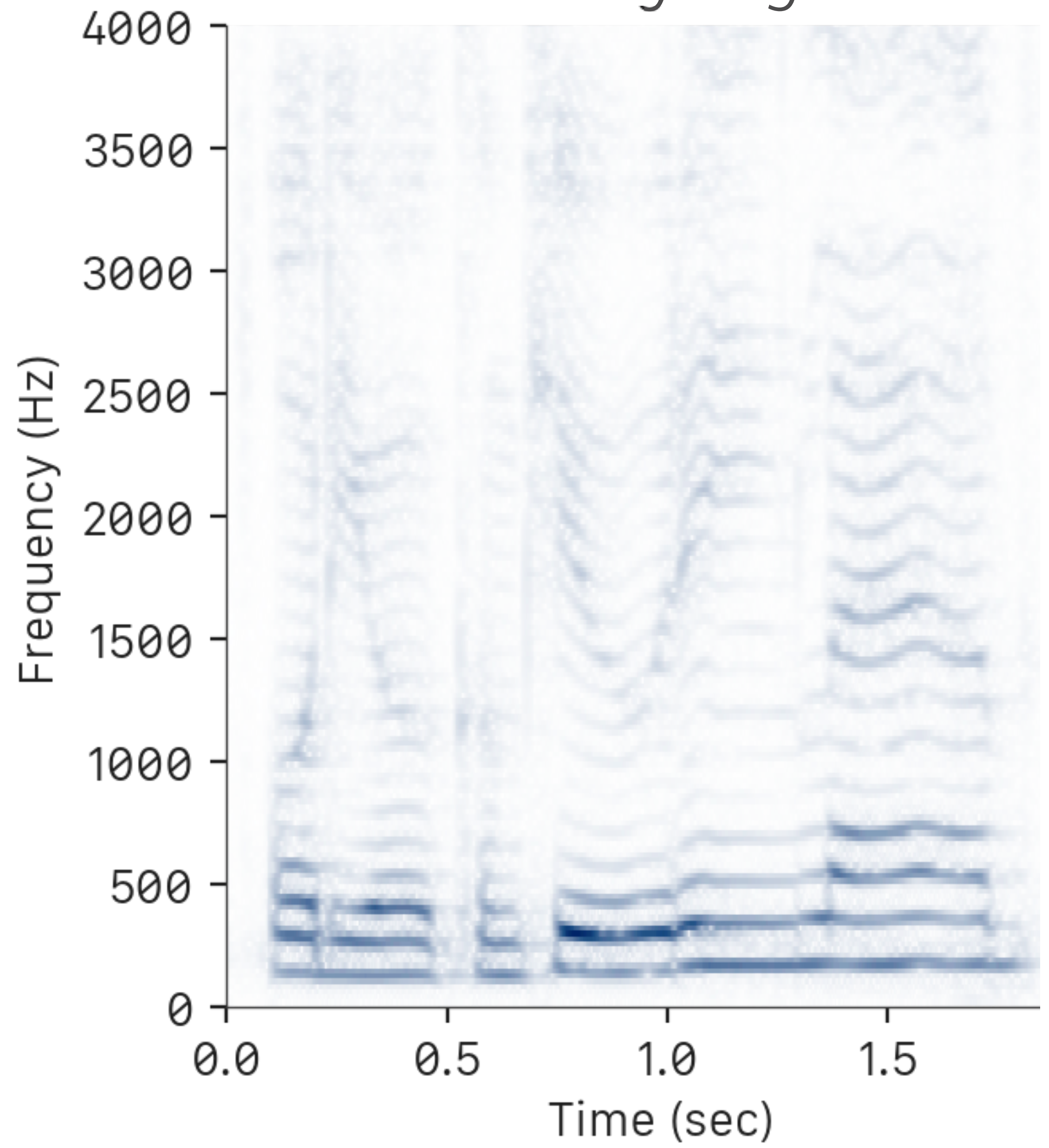


From the frequency perspective

- An (approximately) harmonic sound
  - Has a series of harmonics
  - Fundamental frequency f_0 is the 0-th harmonic
  - Signal can be noisy, or slightly aperiodic, f_0 is not constant

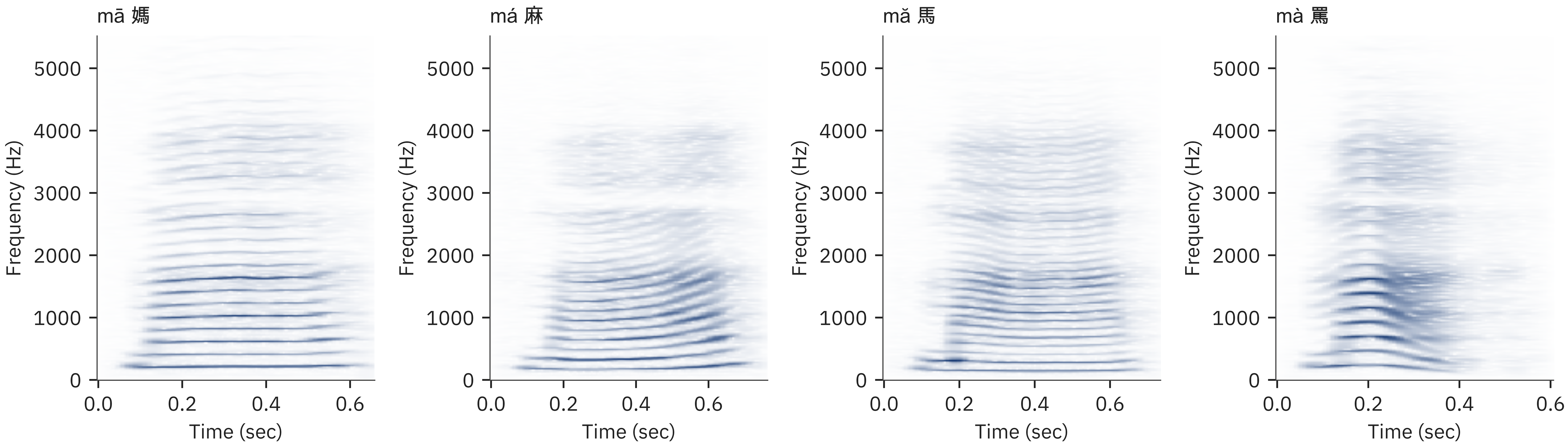


Real-life signal



Why do we care about pitch?

- Speech pitch analysis
  - Measure intonation, prosody, emotion, etc.
    - E.g. identify a question from rising pitch in the end
    - Identify depression, autism, speech impediments, etc.
  - Used for tonal language analysis

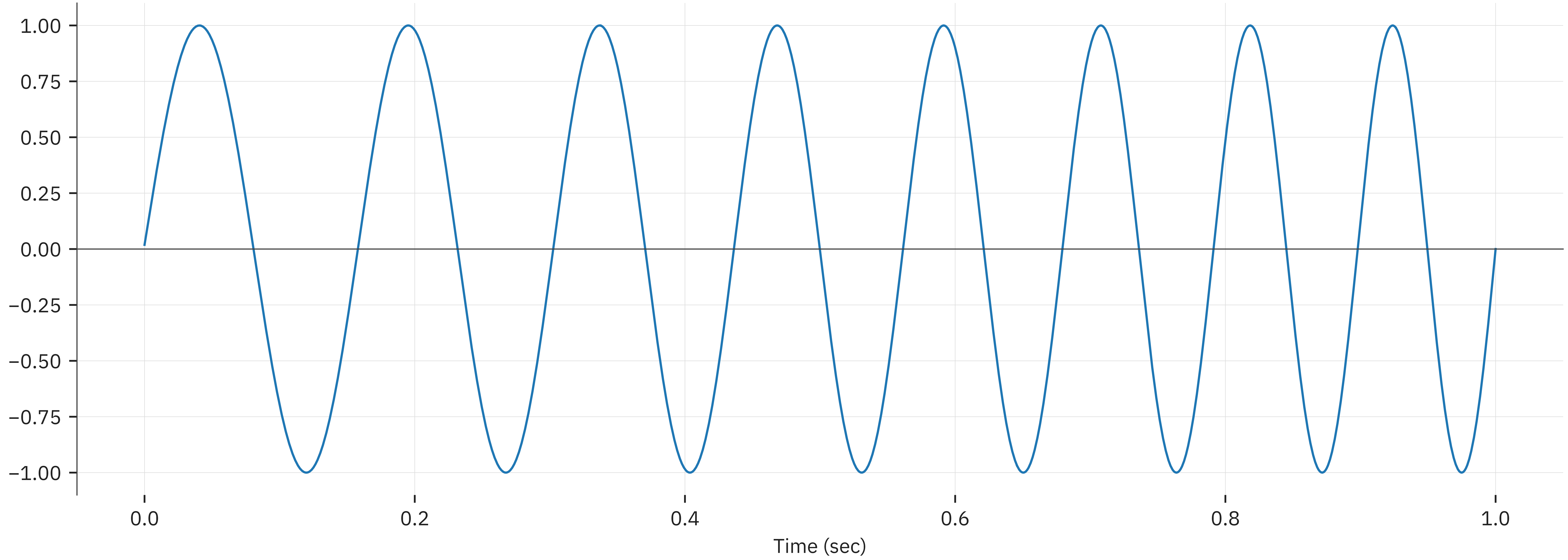


Not just for speech

- Music pitch tracking and correction
- Analyzing historical recordings
  - Music transcription / musicological analysis
  - Performance analysis
- Finding melodies
  - Query by humming
    - Pitch tracking
    - Matching melodies to databases

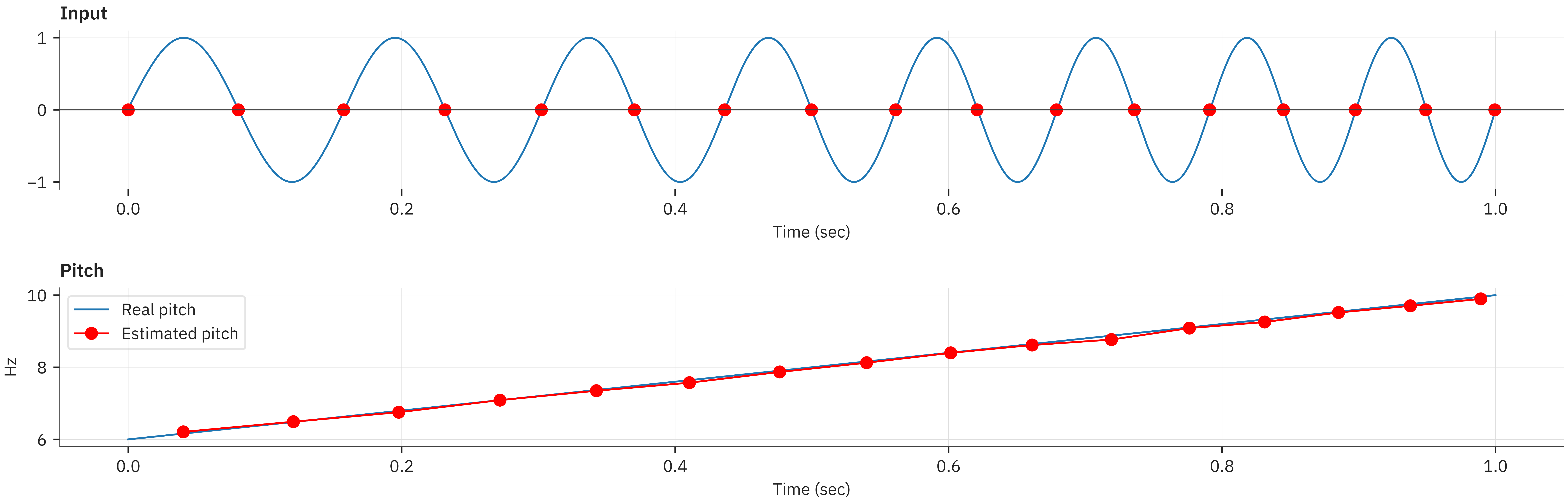
Pitch tracking

- Estimate the value of “pitch” over time
  - Suggestions?



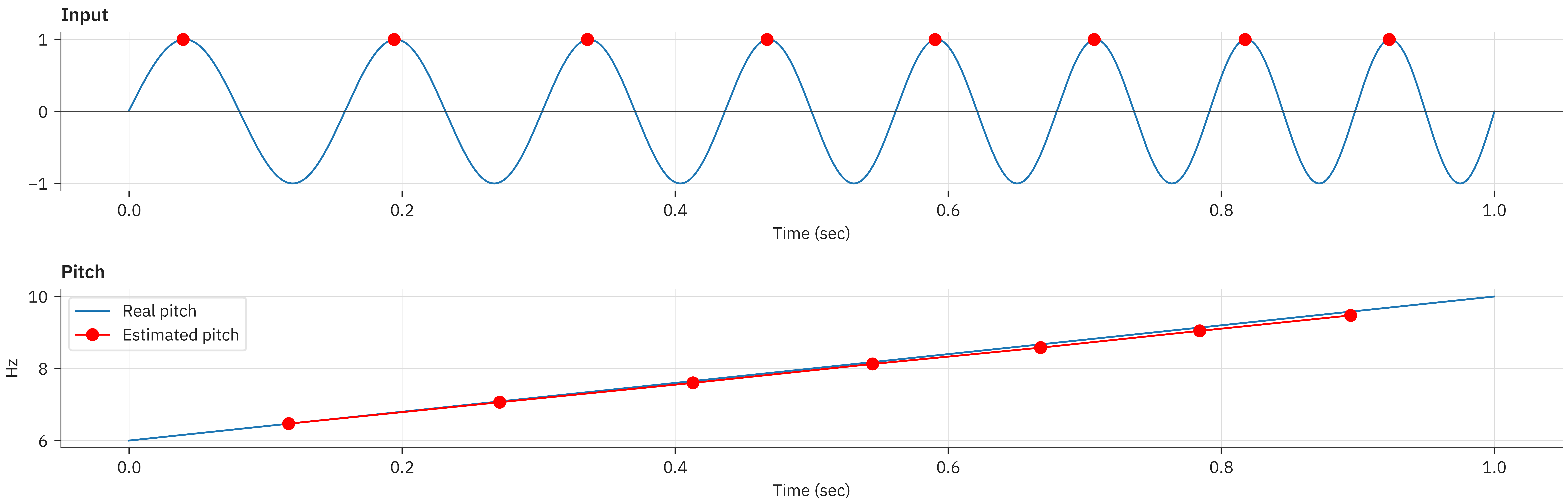
Simple pitch tracking

- The *Zero Crossing Rate* method
  - Find where the waveform crosses the zero mark
    - Usually twice per period: $f = 0.5 \,/\, dz$ *Time between zero-crossings*
  - Very fast to compute, just find $x[t]x[t+1] < 0$



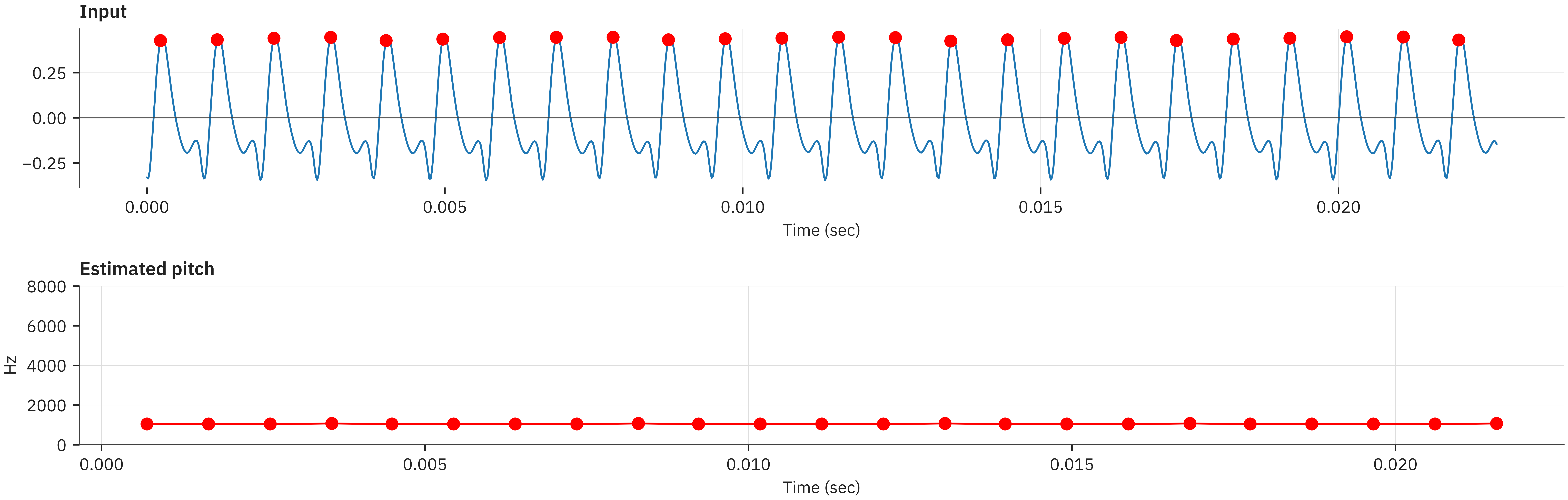
An alternative

- Peak picking method
  - Harder to compute
    - Effectively zero crossings of derivative on positive side
  - More intuitive, but slightly more complex to code



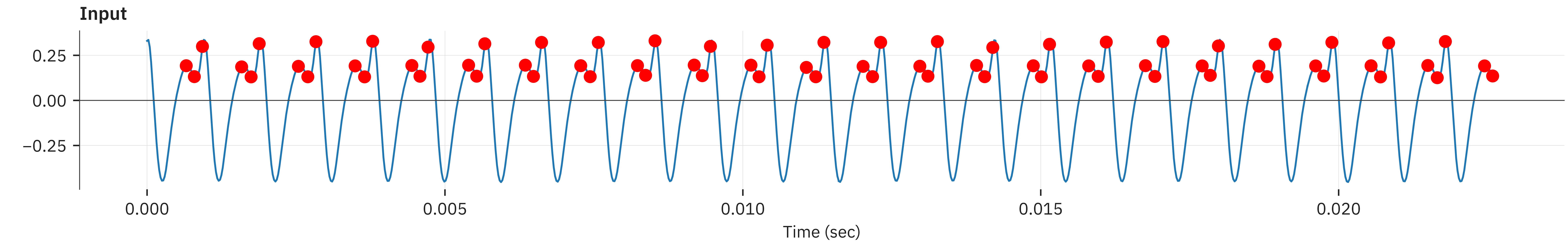
Can work well in many cases

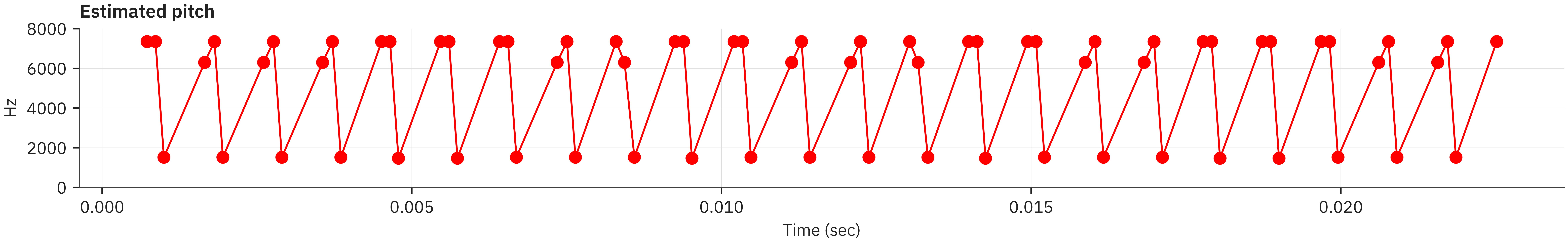
- Trumpet sound
  - Has strong peak at the beginning period
  - Easy to track



But can also fail miserably

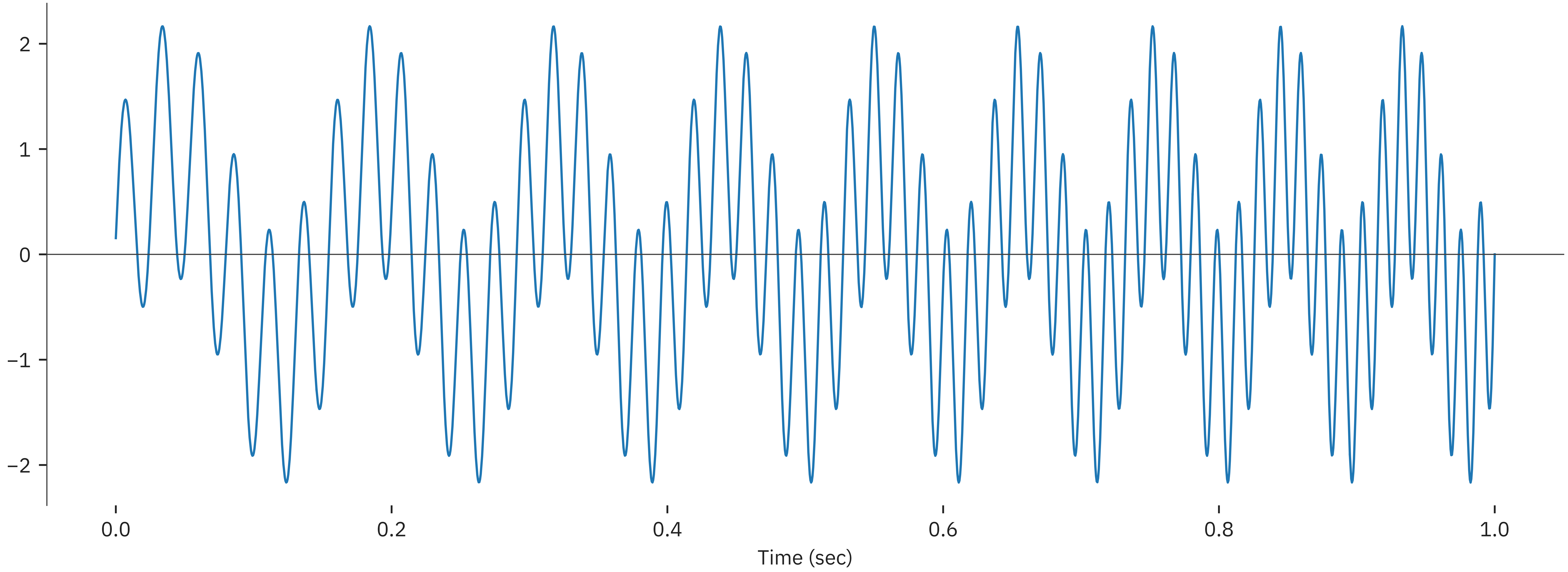
- Same sound, sign-flipped
  - Which peaks are the right ones now?
  - This will not work reliably





Some troublesome issues

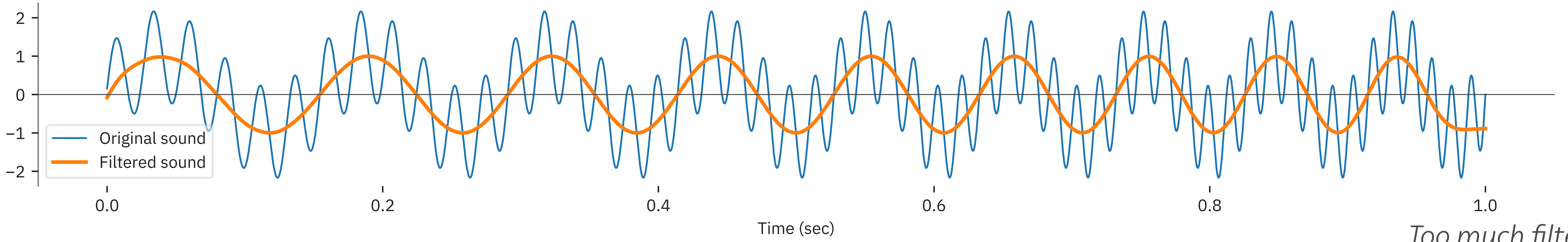
- We can have multiple peaks/zeros in a period
  - In fact, this happens most of the time
  - Is there a simple way to fix that?



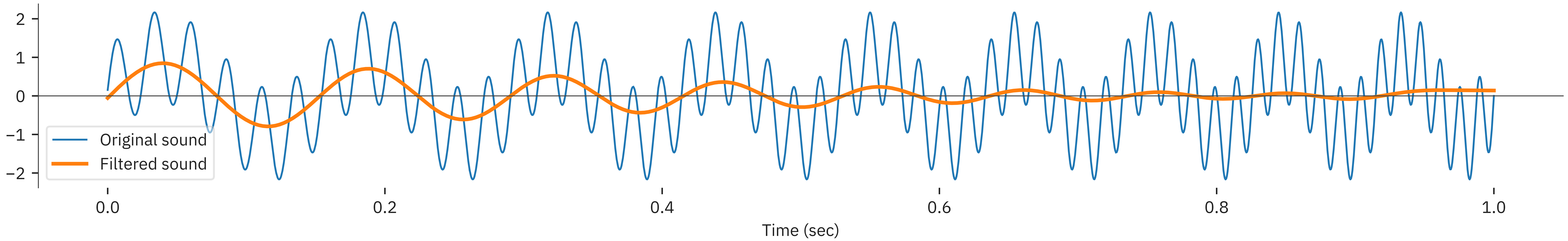
A quick fix

- Apply a lowpass filter to remove excessive wiggling
  - Leaves only the fundamental frequency
    - But might remove that too if it is too aggressive
  - Filter cutoff should be greater than fundamental

Just about right

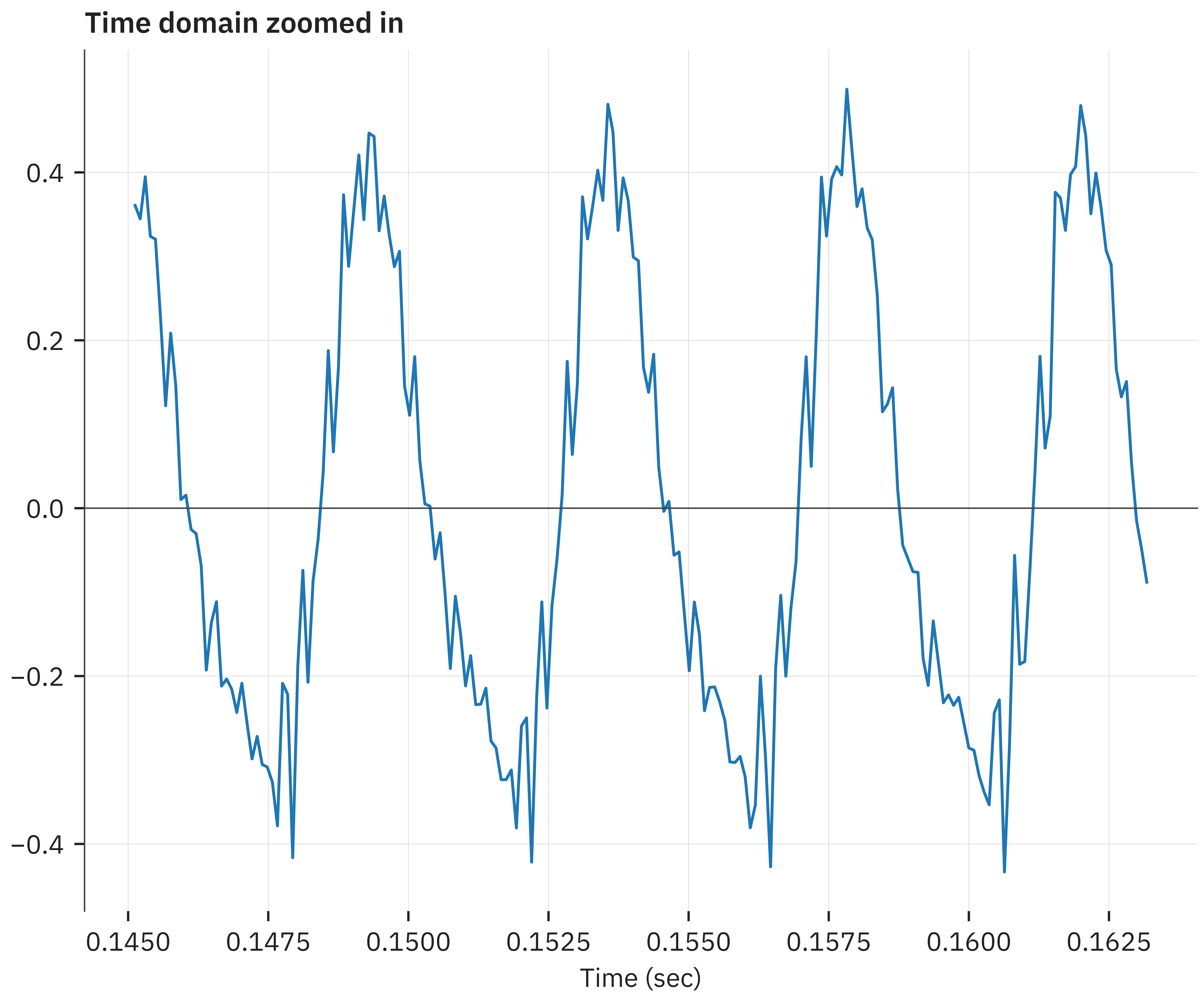


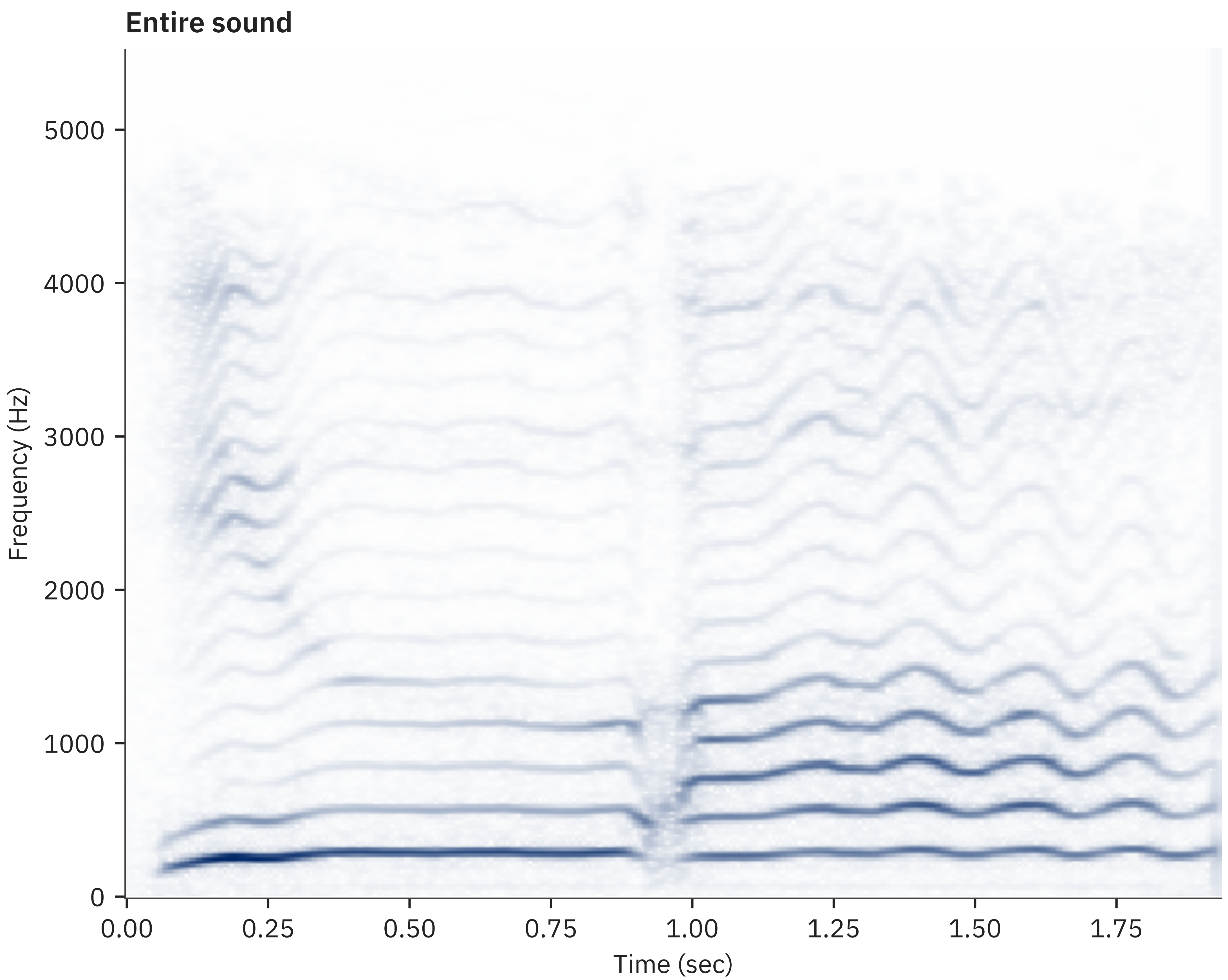
Too much filtering



On a real-life sound

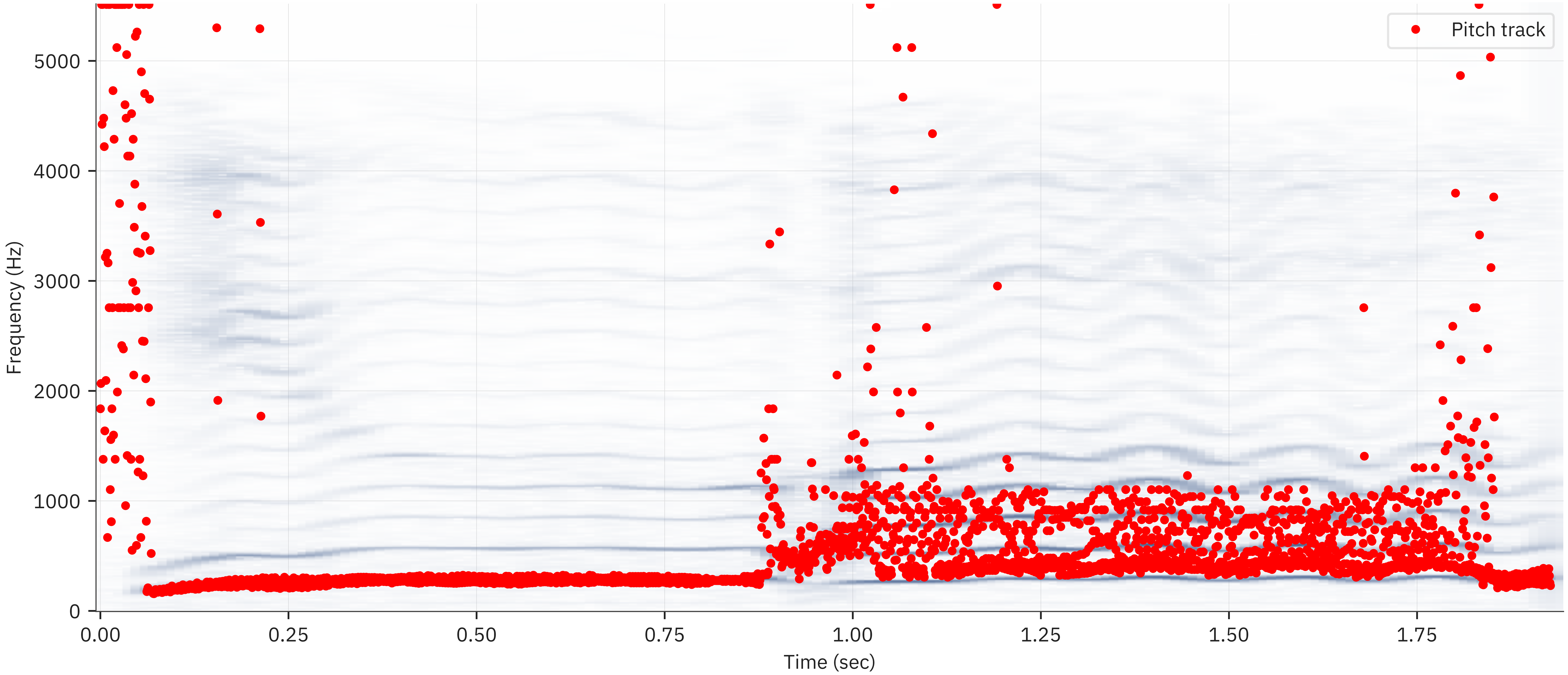
- Example singing sound
  - No clear zeros or peaks





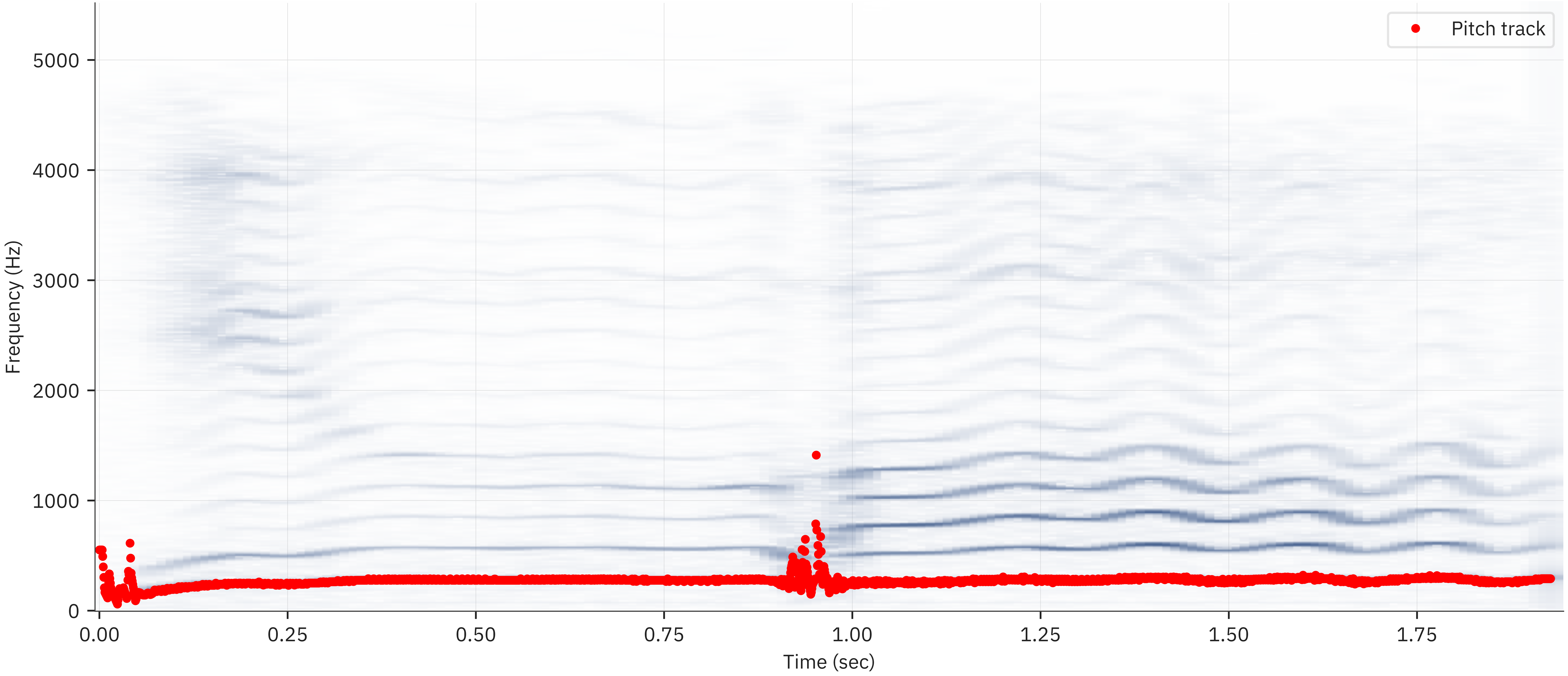
Using zero crossings

- Sometimes works, but not always
  - Lots of bad estimates in the second half



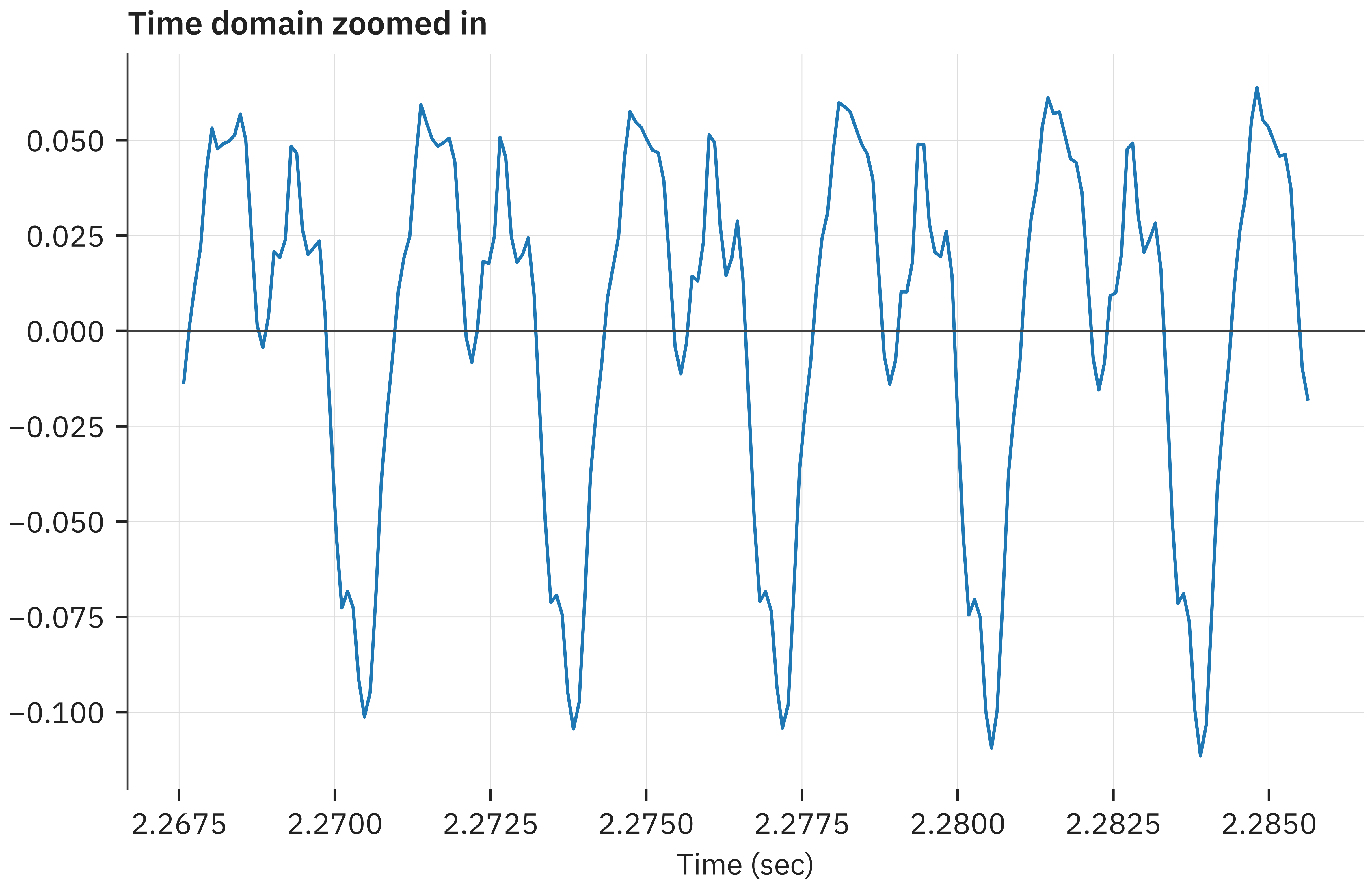
After some filtering

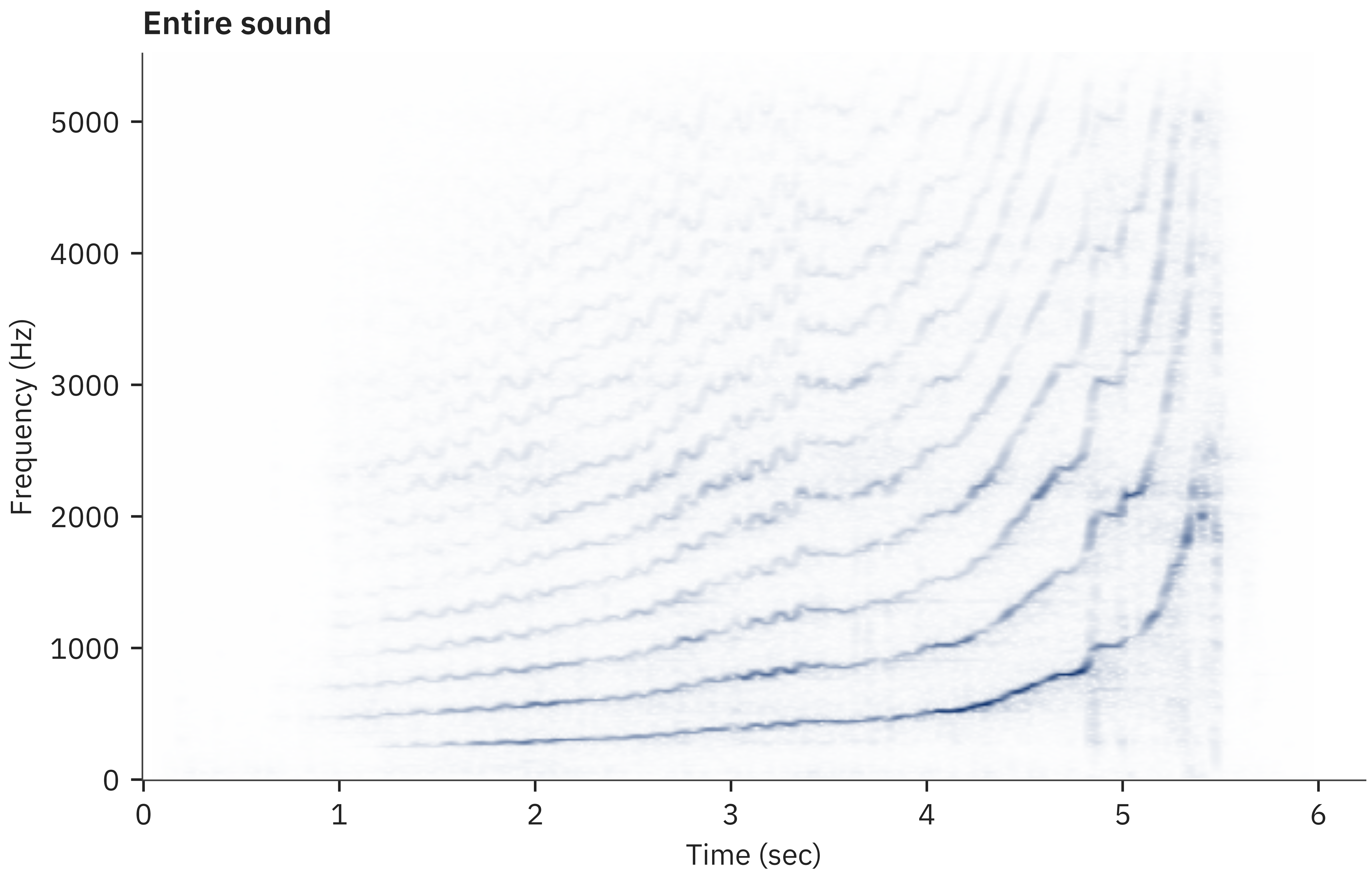
- Much smoother pitch tracking
  - Not as good at unpitched sections though (that's ok)



One more nasty case

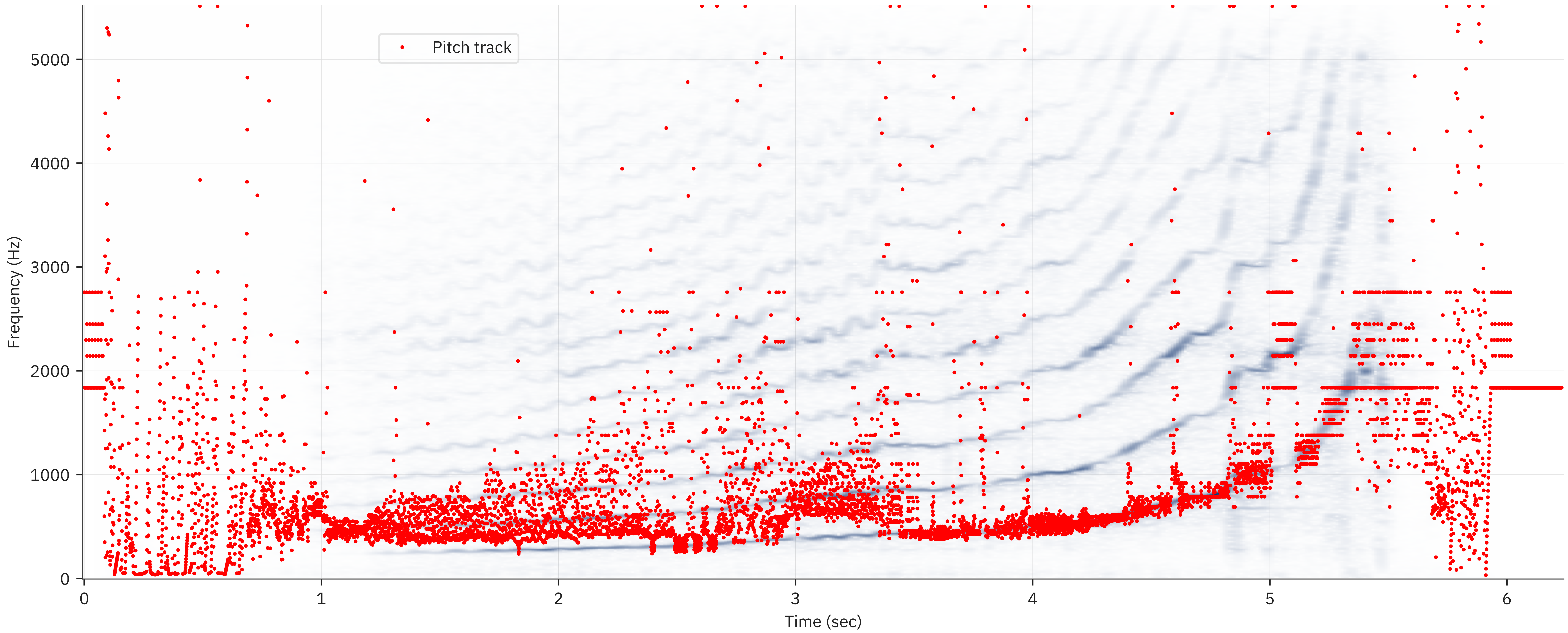
- Violin glissando
  - More challenging case
  - Fundamental goes very high (so we can't lowpass!)





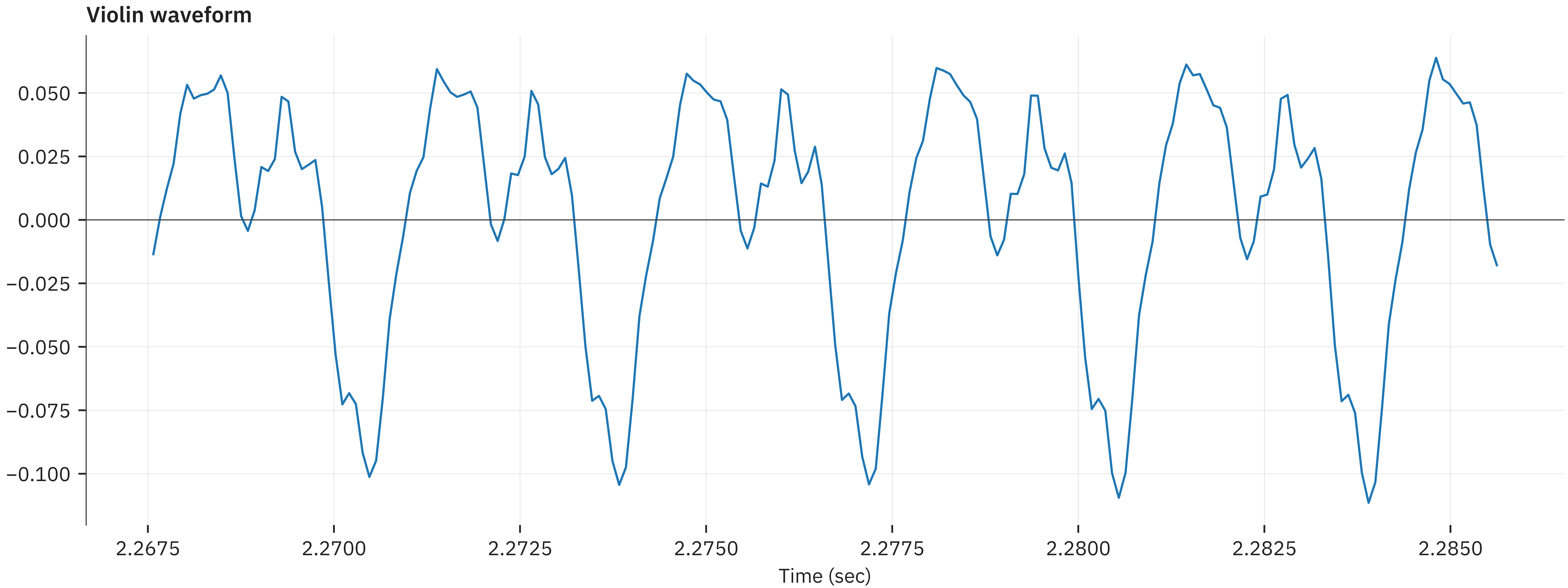
Result

- Very noisy pitch estimates
  - Lots of mistakes despite smoothing



Back to the drawing board

- What else can we measure to get a pitch?
  - How about periodic similarity?



Measuring waveform similarity

- *Cross-correlation*:

$$c_{x,y}[k] = \sum_t x[t]\, y[t+k]$$

  - Measures waveform similarity at offset of k samples
    - Looks familiar?
- *Auto-correlation*:

$$c_{x,x}[k] = \sum_t x[t]\, x[t+k]$$

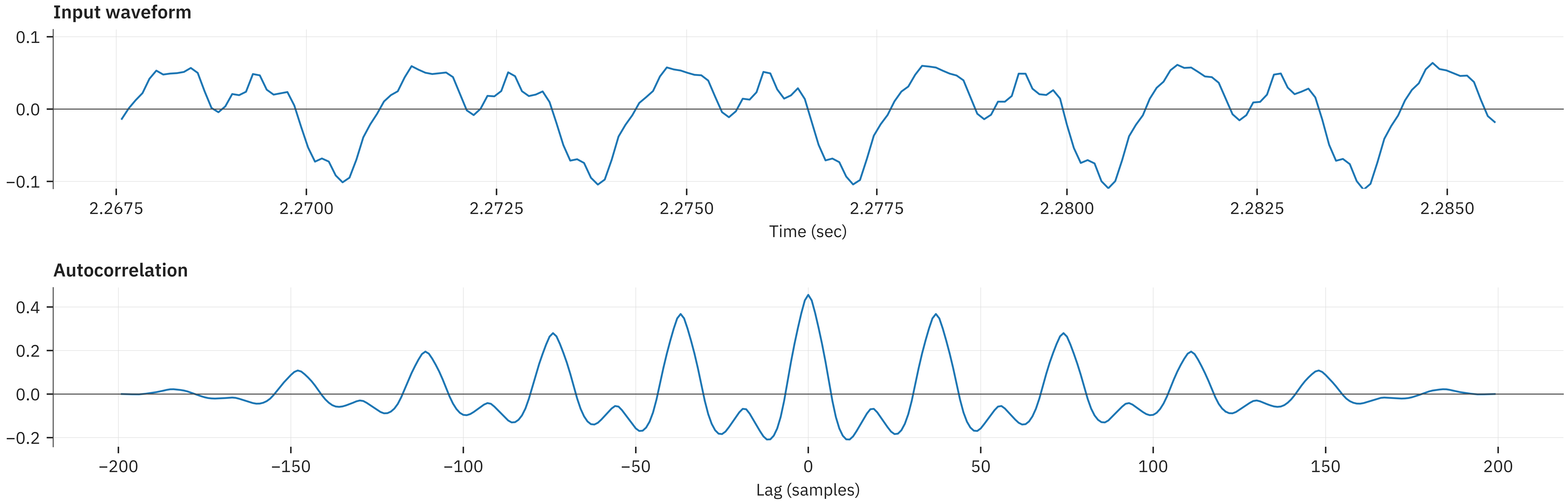
  - Measures similarity of signal with itself shifted by k samples

Measuring periodic similarity

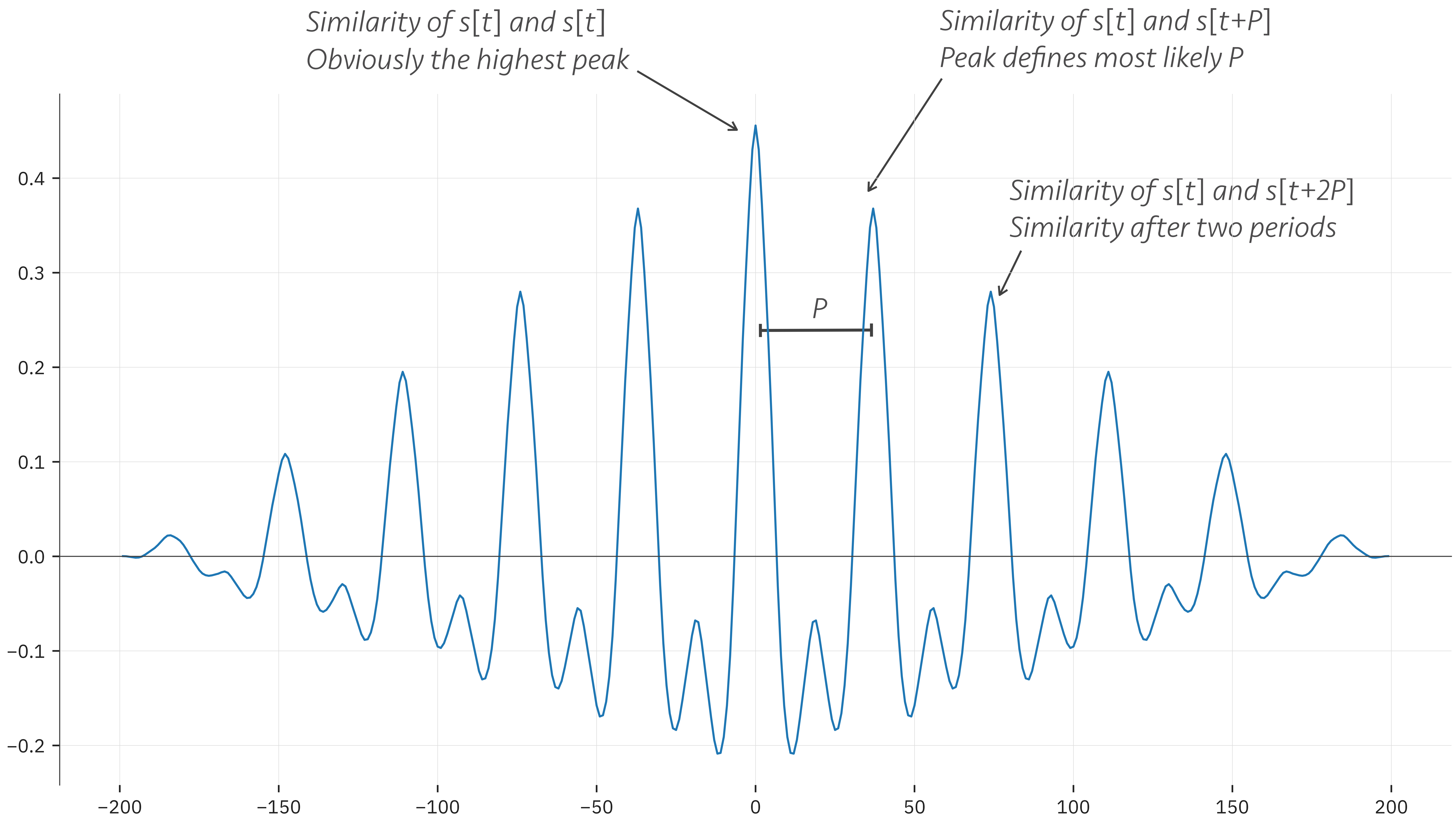
- Assumption: $s[t+P] = s[t] + n[t]$

  Little bit of noise since waveform won't be perfectly periodic

  - So $c_{x,x}(P)$ should be high
    - Because of periodicity
    - What other values will be high?



Interpreting the autocorrelation



One more trick

- Why not use the FFT to make this faster?
  - Cross-correlation is convolution with one of the operants flipped in time
  - Convolution is: $x * y = \mathrm{DFT}^{-1}\Big(X[\omega]Y[\omega]\Big)$
  - Flipping in time is: $x[-t] = \mathrm{DFT}^{-1}\Big(X[\omega]^*\Big)$
  - Cross-correlation is: $c_{x,y} = \mathrm{DFT}^{-1}\Big(X[\omega]^*Y[\omega]\Big)$

Pitch tracking with autocorrelation

- Perform STFT
  - For each time frame compute autocorrelation:

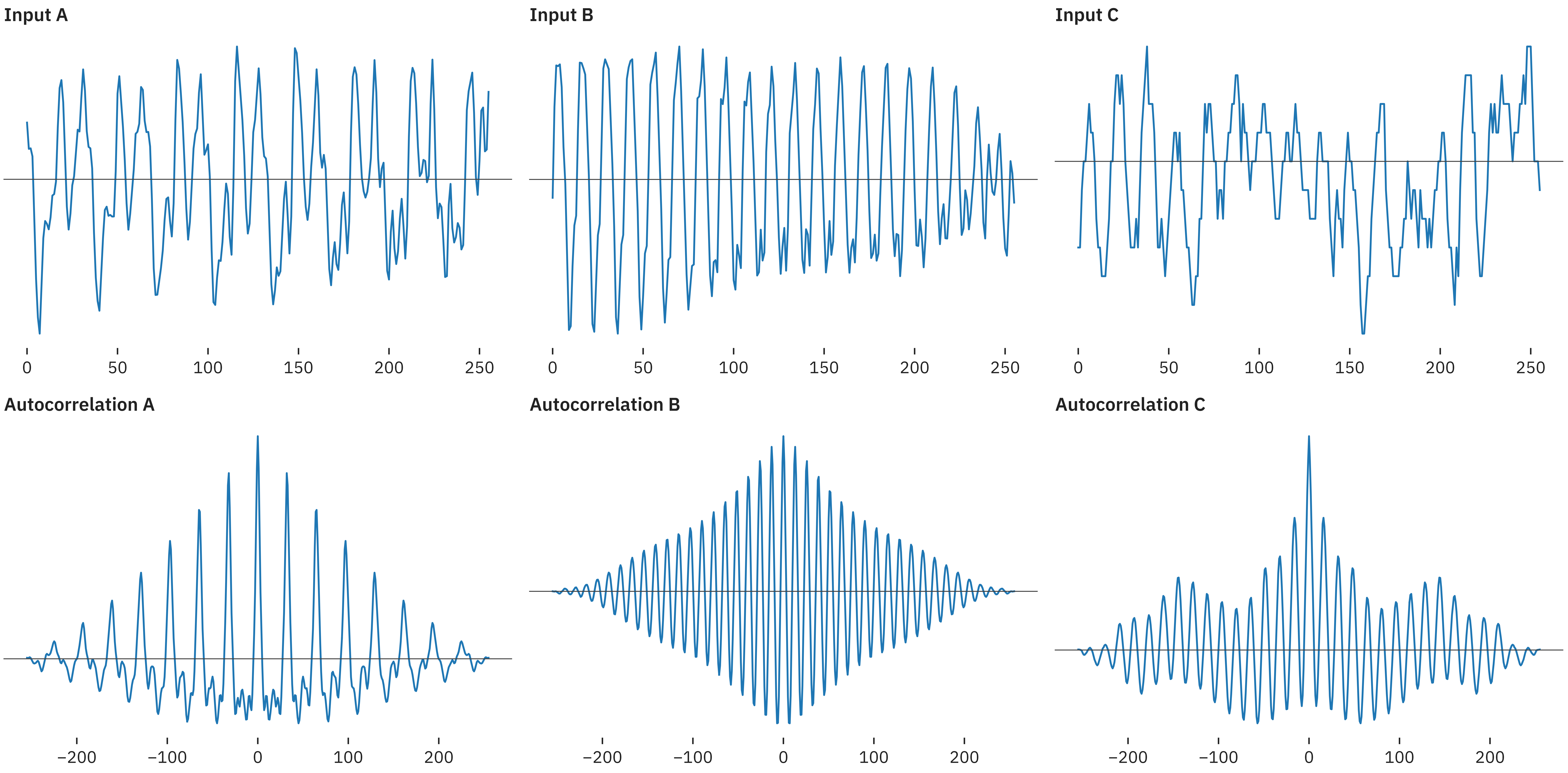
$$C_\tau[\omega] = X_\tau[\omega]^* X_\tau[\omega]$$

  - Take inverse DFT to obtain time version:

$$c_\tau = \mathrm{DFT}^{-1}\Big(C_\tau[\omega]\Big)$$

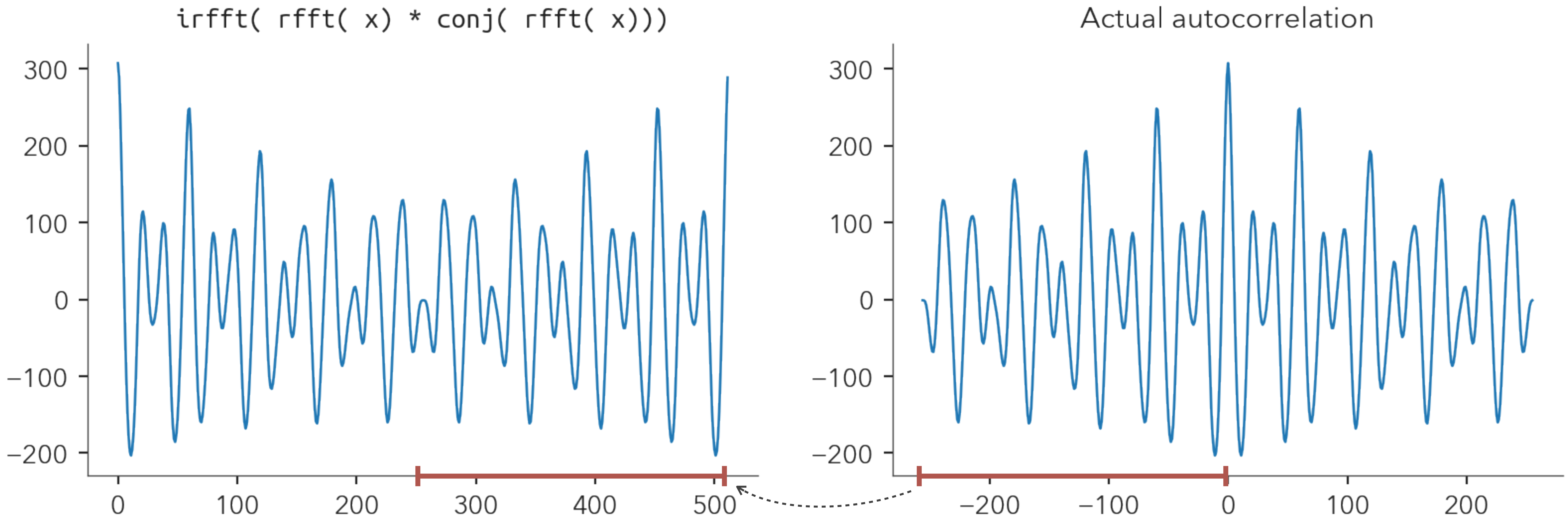
  - Get first significant peak to find current period
    - Significant = above ~80% of main peak

Example autocorrelations



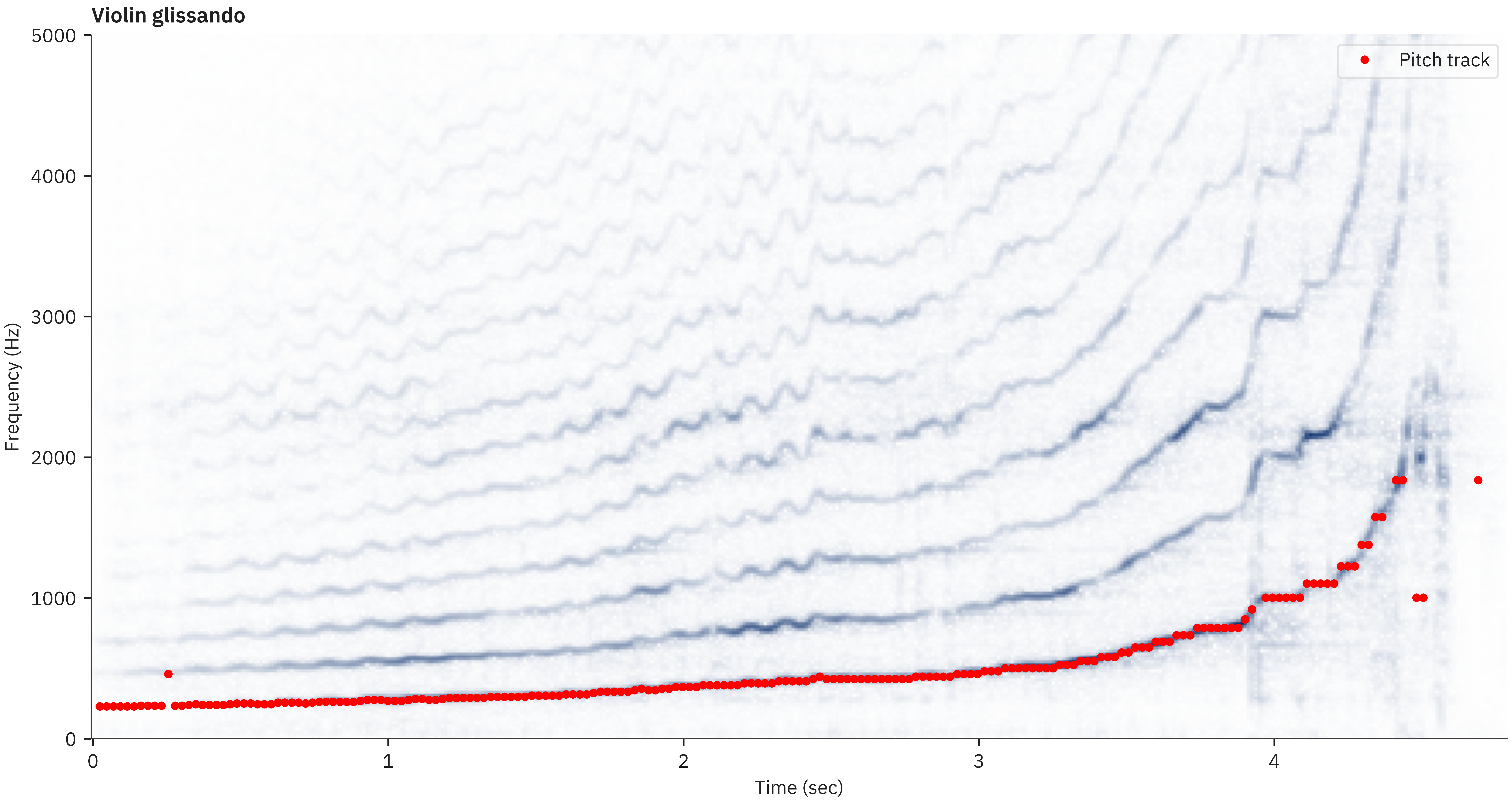
Practical word of caution

- When coding this, it will come out different:
  - `c = irfft( rfft( x) * conj( rfft( x)))`
  - The negative shift correlation values will be the second half



Back to the violin example

- Much better estimate than before

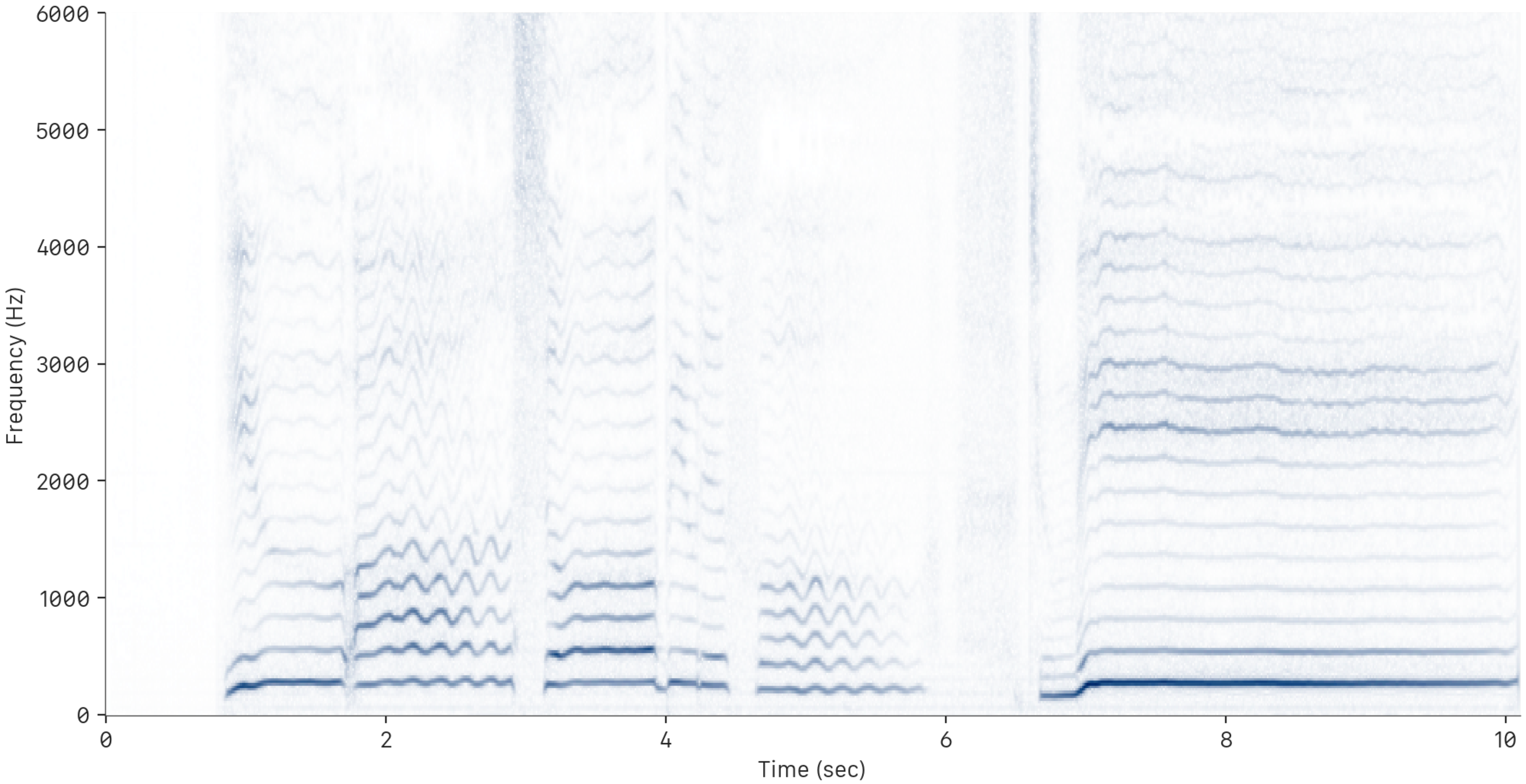


Some tricks

- Smoothing out occasional jumps
  - Often we see octave jumps, which are mistakes
  - We can use a median filter on the pitch estimate to smooth it
- Imposing limits on the period size
  - Cannot be too small (= high pitch)
  - Cannot be too low (= spurious peak)
- Periodicity should be strong
  - Height of autocorrelation peak should be high
    - If not, it means that we have weak periodicity
      - Either a spurious peak, or the signal is unpitched

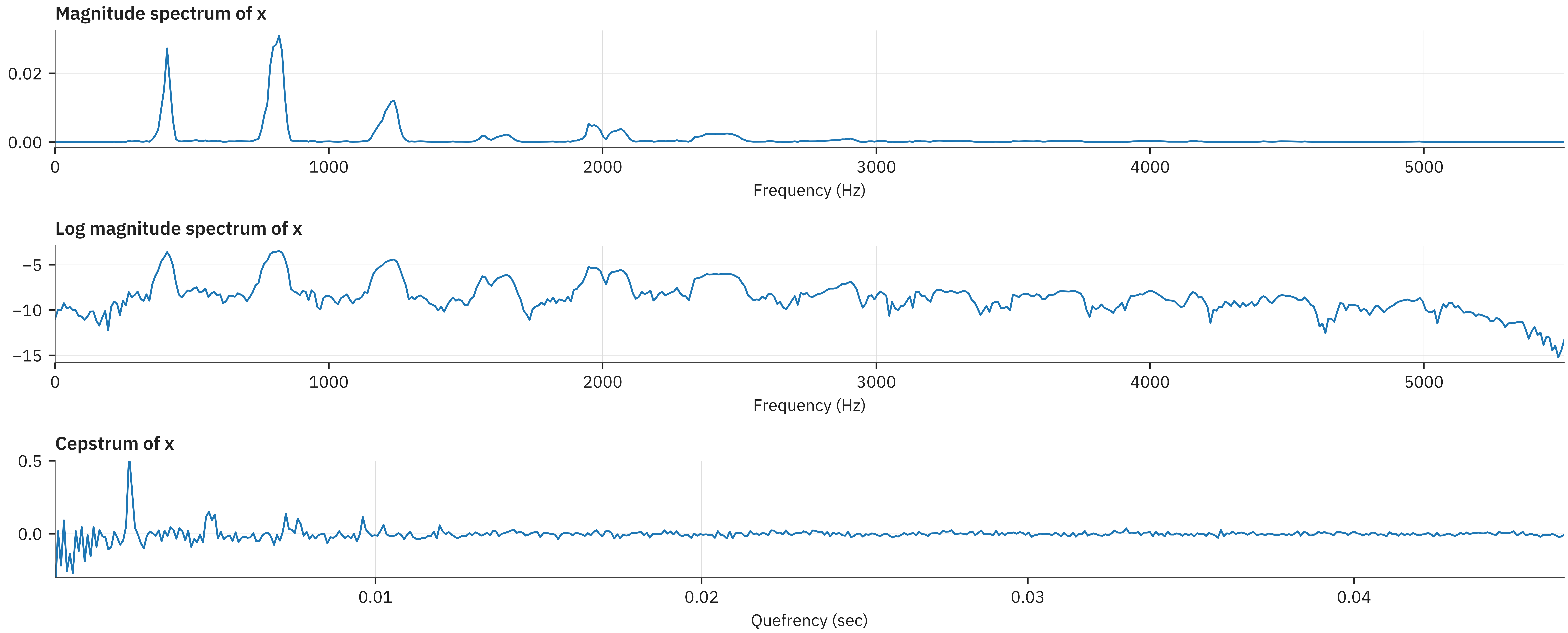
Frequency domain approaches

- Use peak-picking on spectra
  - Discover harmonic series over frequency
    - Can get tricky …



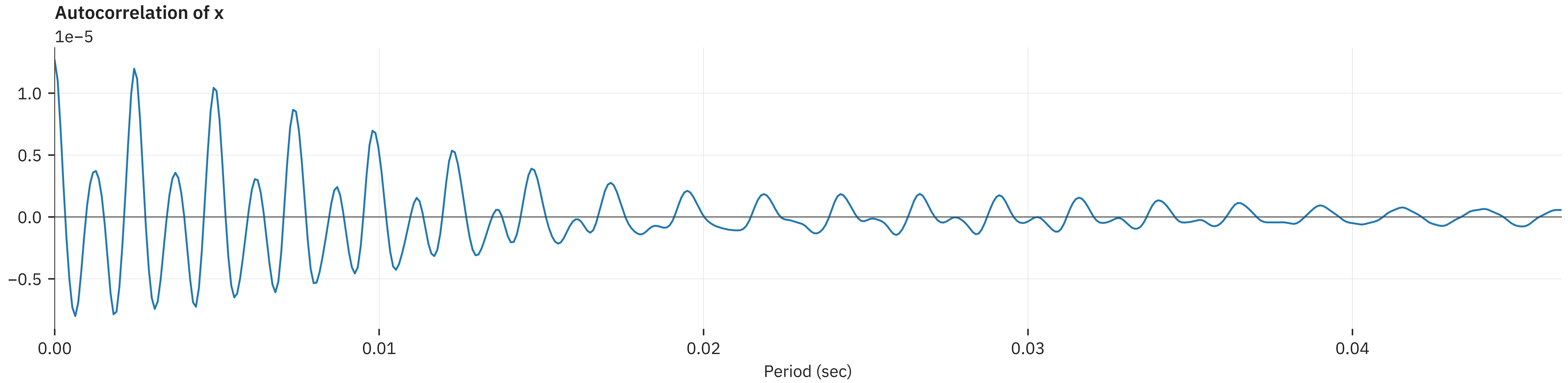
Using the Cepstrum

- The *cepstrum*: $X_\tau[q] = \mathrm{DFT}^{-1}\left(\log|X_\tau[\omega]|\right)$



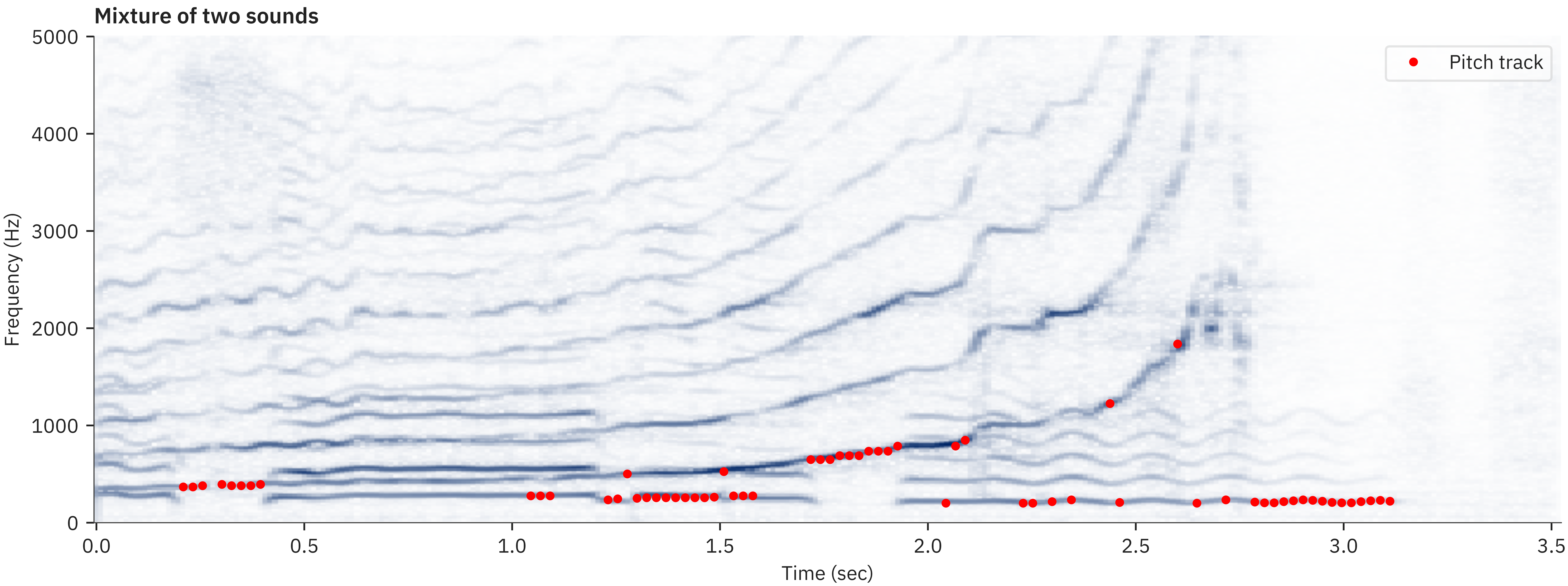
Cepstrum vs. Autocorrelation

- The cepstrum exhibits stronger peaks
  - A little easier to detect (but the code needed is about the same)



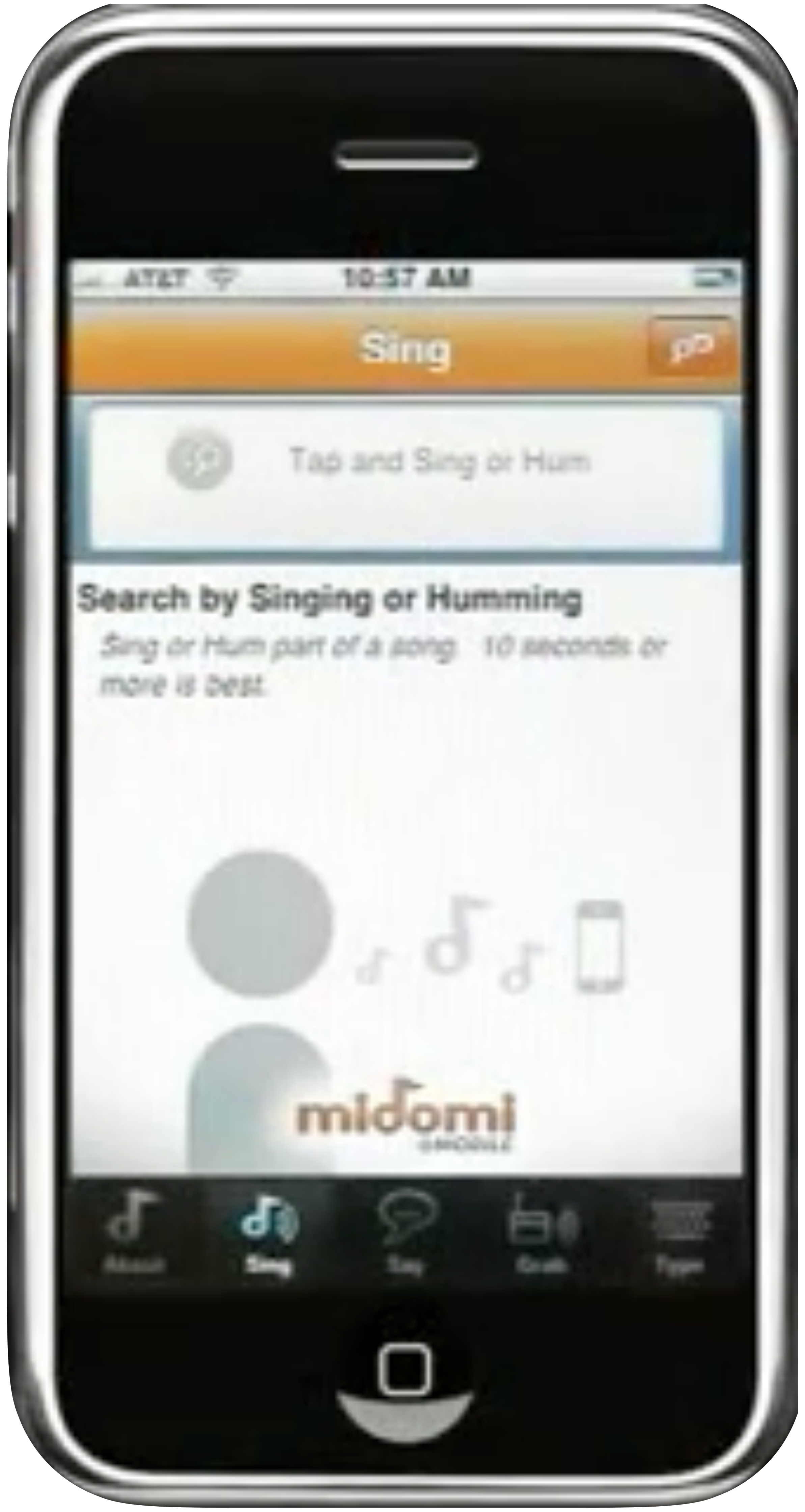
Trouble with polyphonic inputs

- What happens when we have two pitches?
  - E.g. an instrument duet?



Neat applications of pitch tracking

- Query by humming
  - Search a song by singing it
- Pitch track user input and match to song database
- Can get tricky
  - Most people cannot really sing a song in tune!



Neat applications of pitch tracking

- Score following / page turning
  - Determine where in a music score the player is and then turn the score page appropriately

Pitch shifting

- Singing voice input

- Tasks:
  - Shift pitch up by a factor of 1.5
  - Shift pitch down by a factor of 1.5

- How do we do this?
  - Suggestions?

Simple (but wrong)

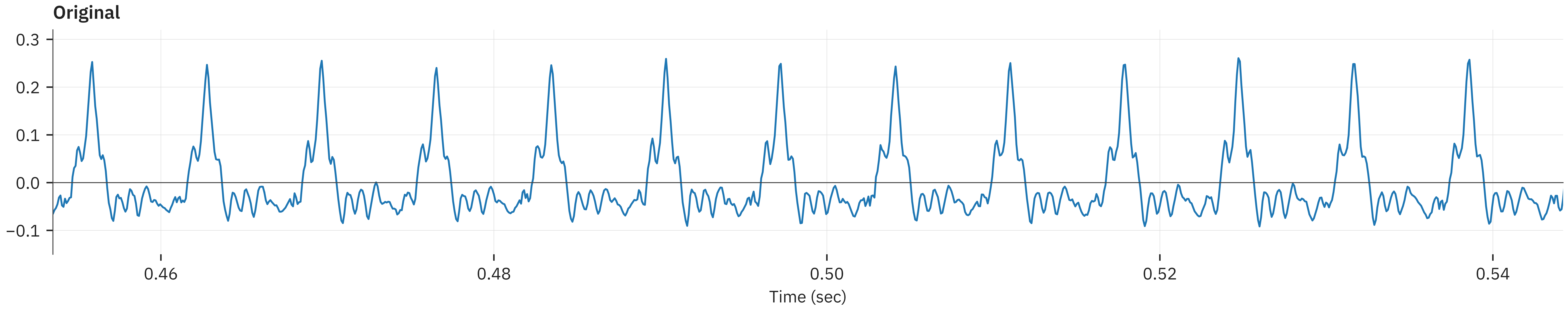
- Change the sample rate!
  - Play faster == pitch goes up
  - Play slower == pitch goes down
- Problems with this idea
  - Sound lasts longer/shorter
  - Sounds like a chipmunk or an ogre
- What's happening here?

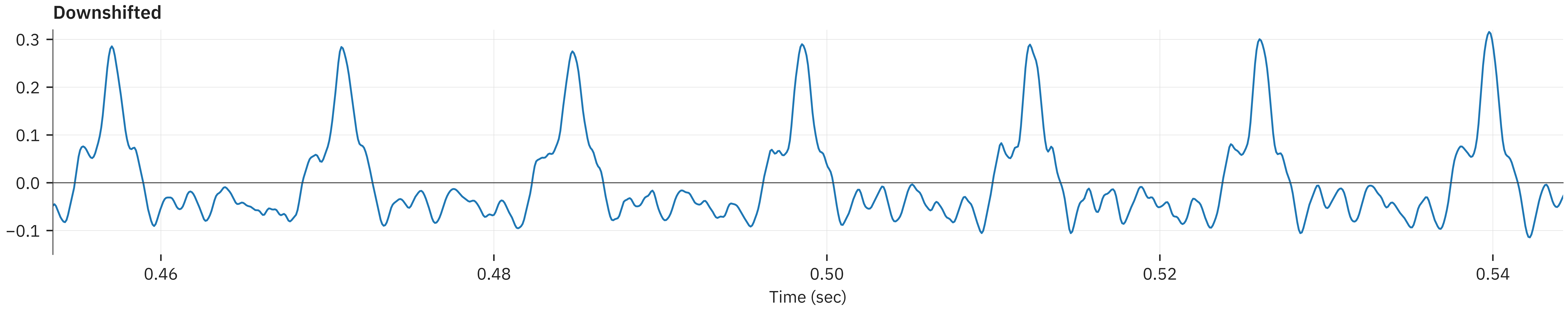
Faster = higher pitch, shorter sound

Slower = lower pitch, longer sound

We do not preserve the wave shape

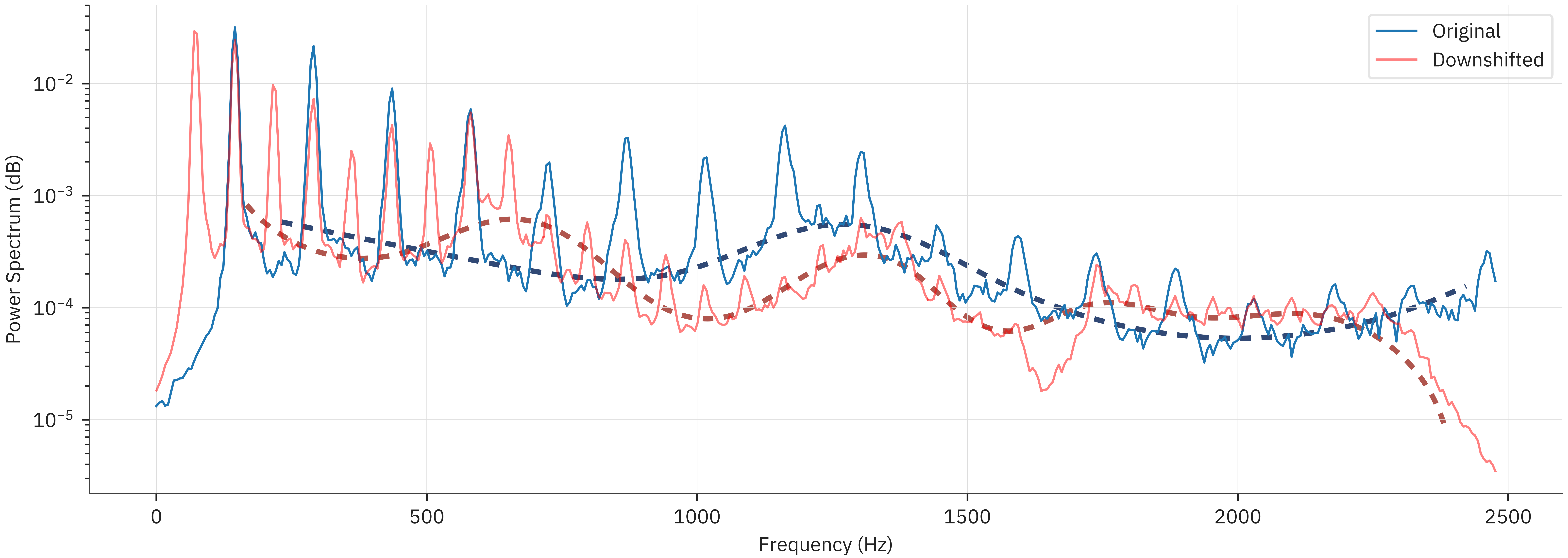
- We should not change the shape of the waveform periods, only the spacing between them





Misplacing the formants

- We only want to shift the harmonics, not the *formants* (resonances) of the speech signal



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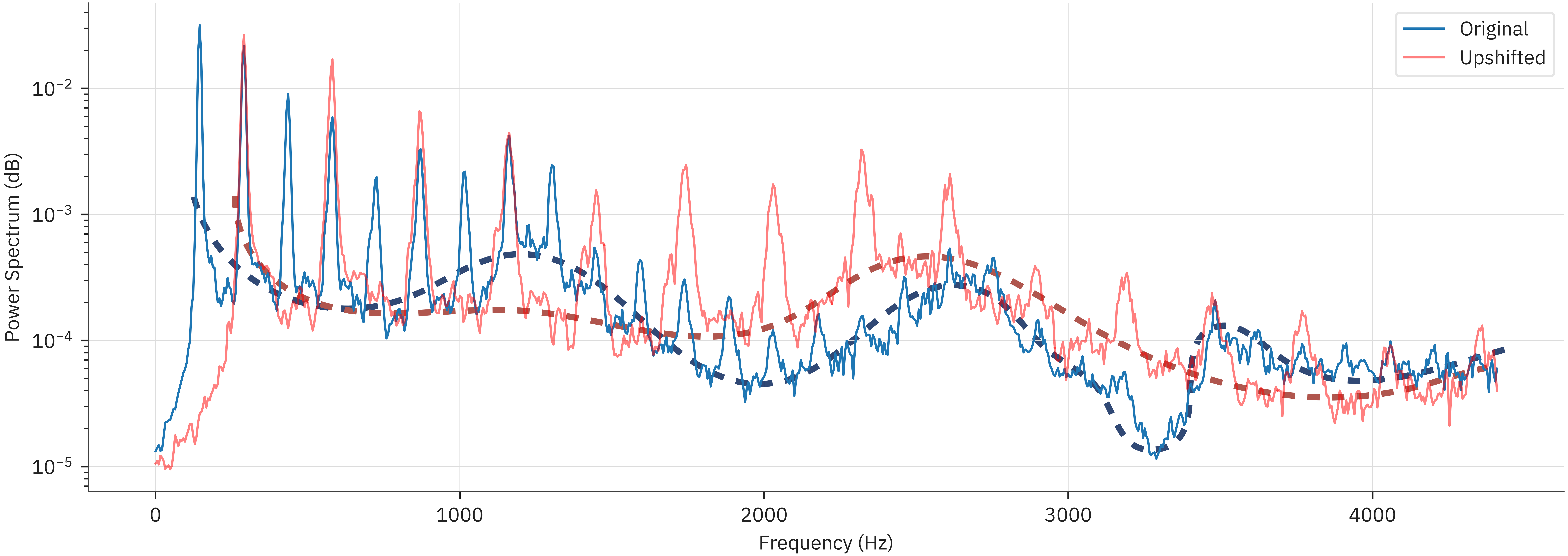
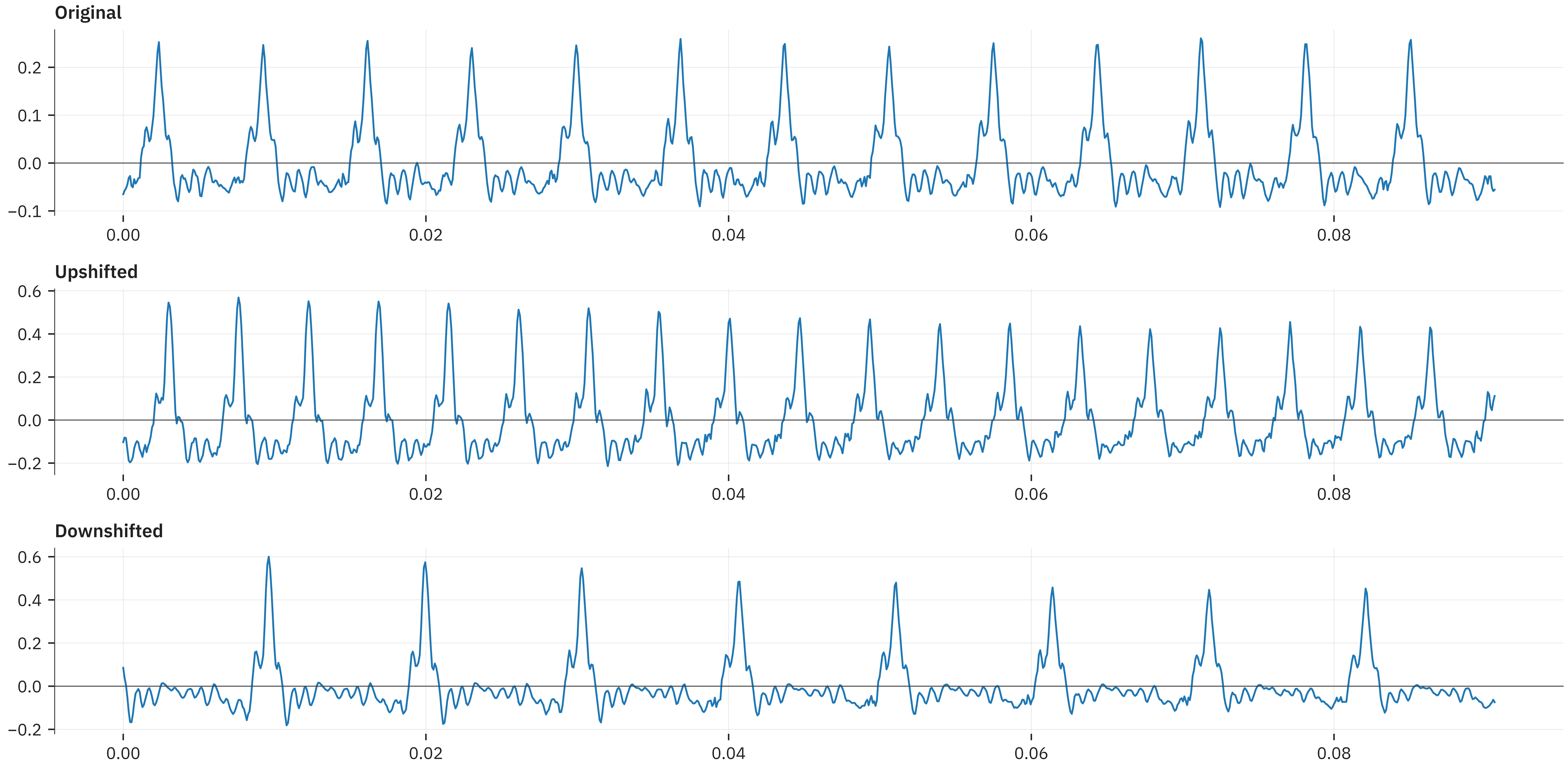


PSOLA

- *Pitch synchronous overlap add*
  - Find the periods of the input signal
  - Space them farther apart, or closer, to alter pitch
    - But maintain their shape!
- Key component
  - Find pitch periods (we know how to do that!)
- Once found, reposition and overlap-add them
  - Reposition to change periodicity

Proper shifting

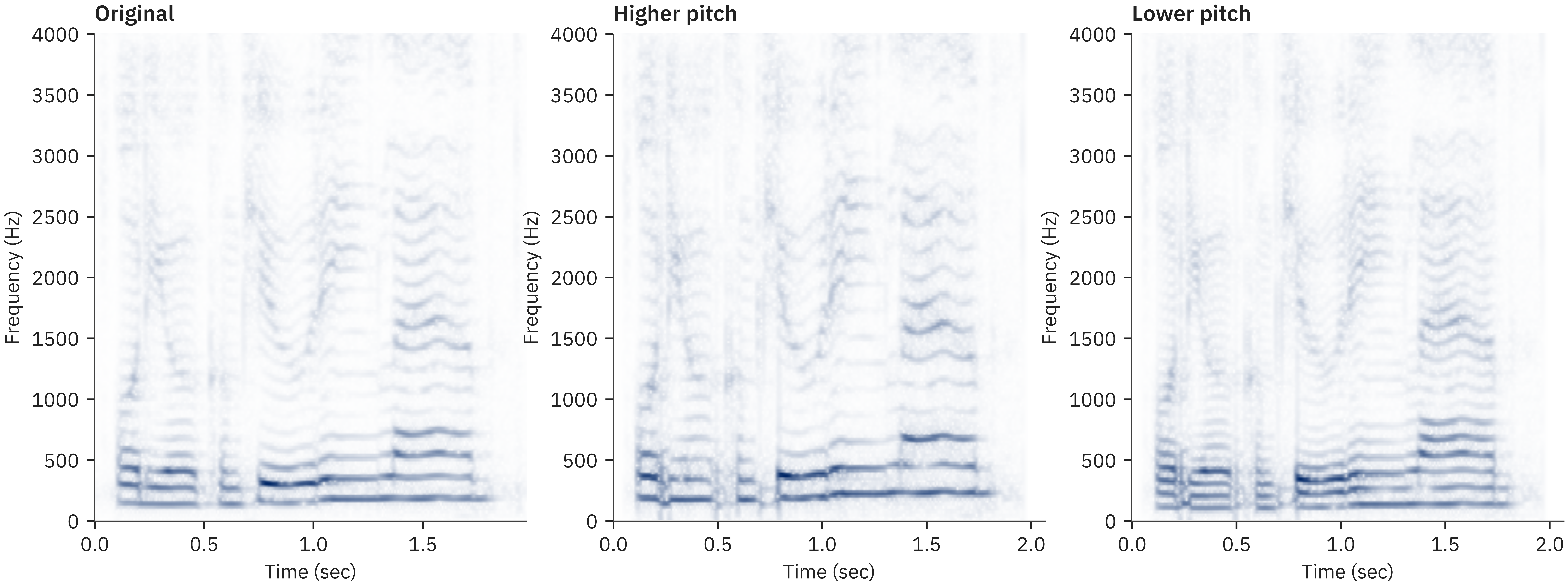
- Note that the characteristic period shape is the same, and isn't stretched out



Some examples

- Works well for pitched speech
  - But we cannot push it to extremes

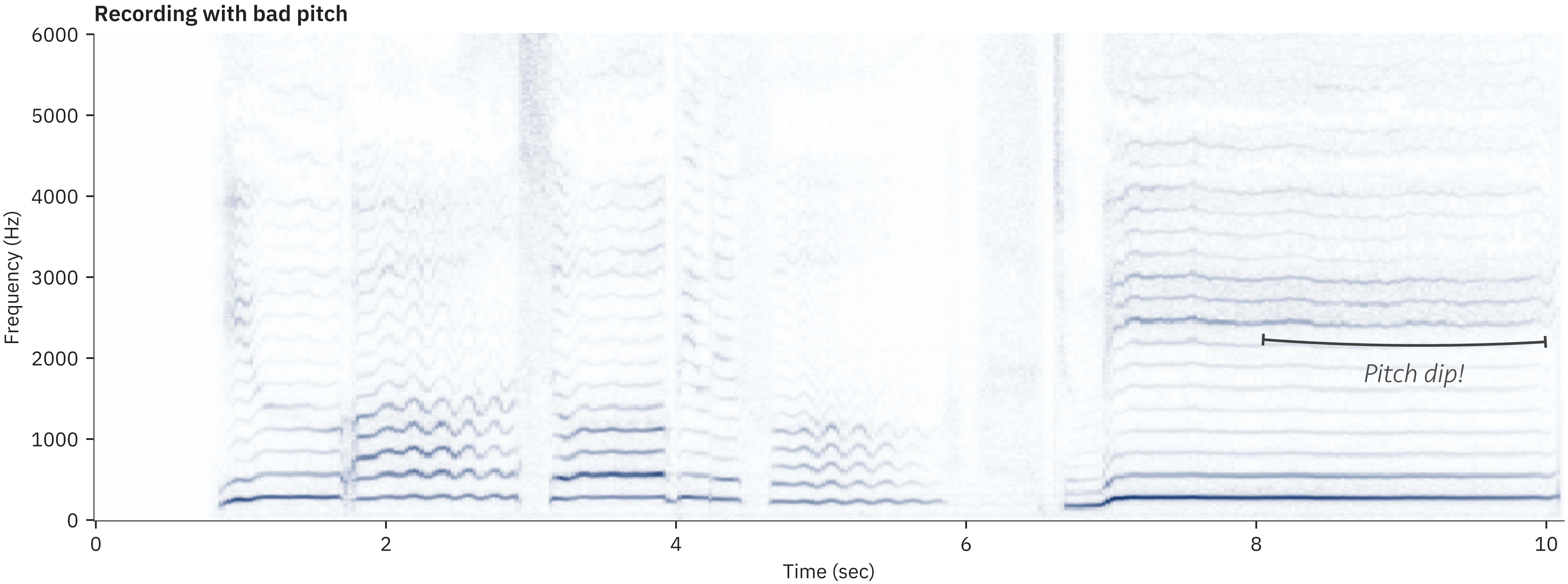
Too high *Too low*



All!

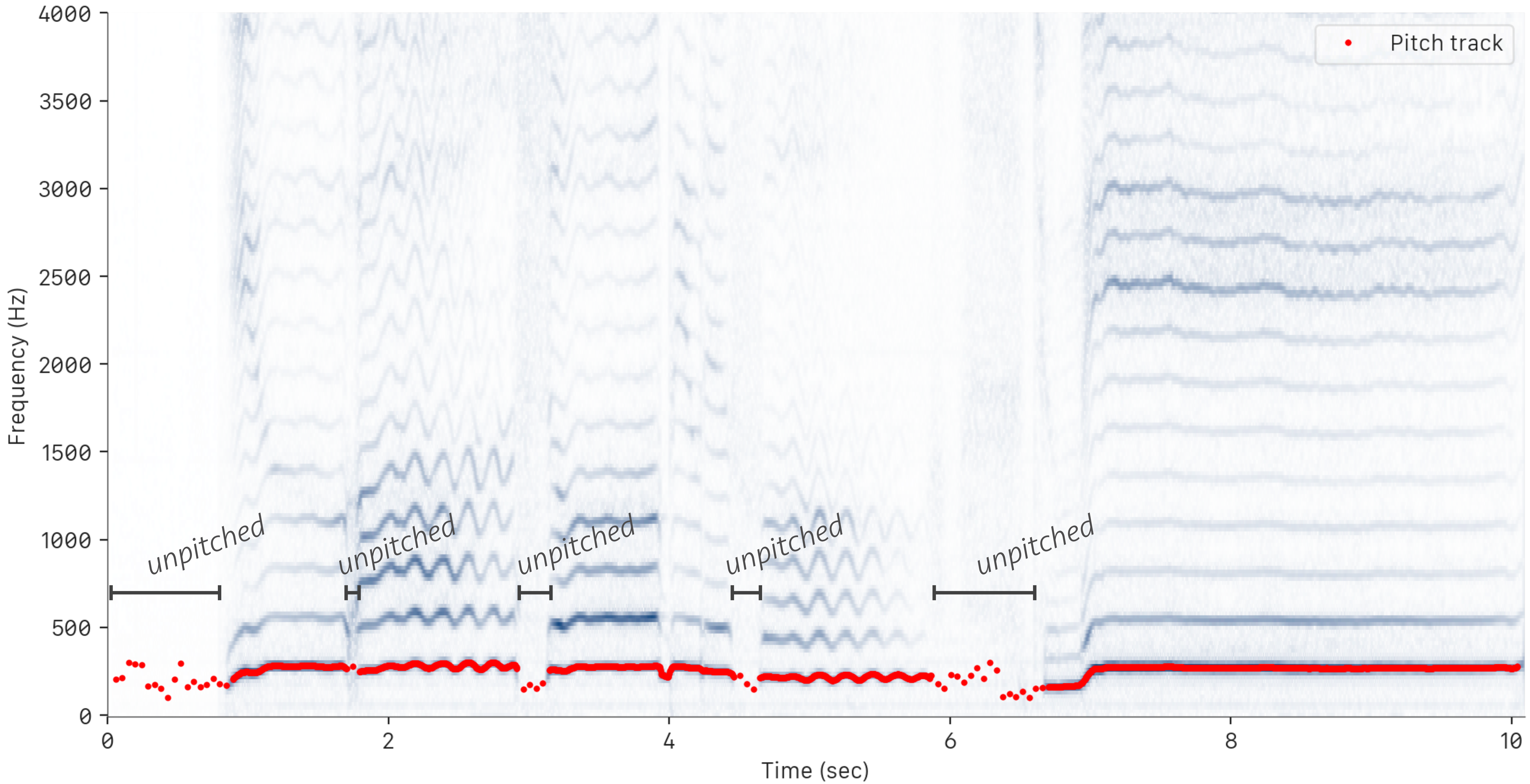
Autotuning

- Combining pitch tracking and PSOLA
  - A motivating example



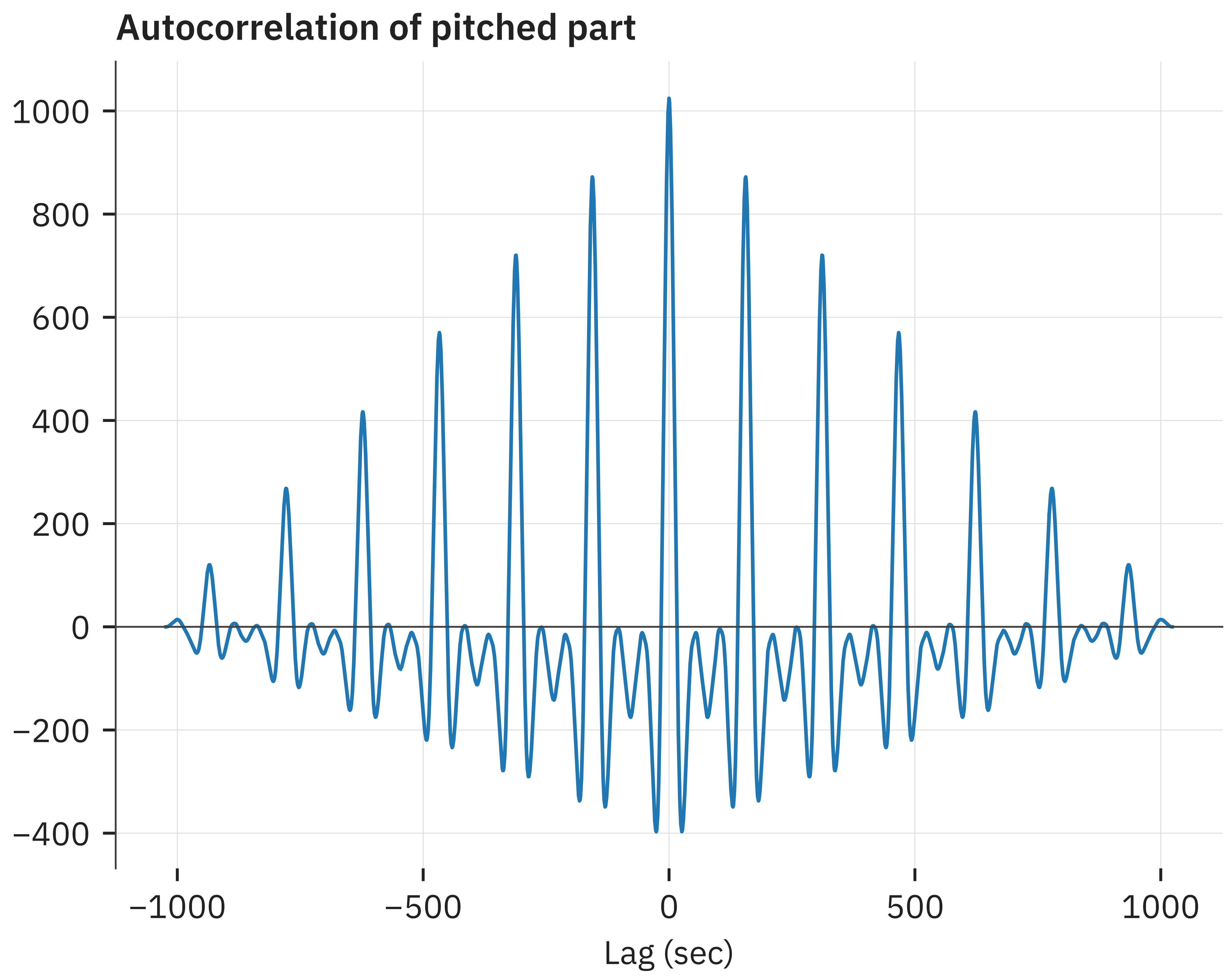
Getting the sung pitch

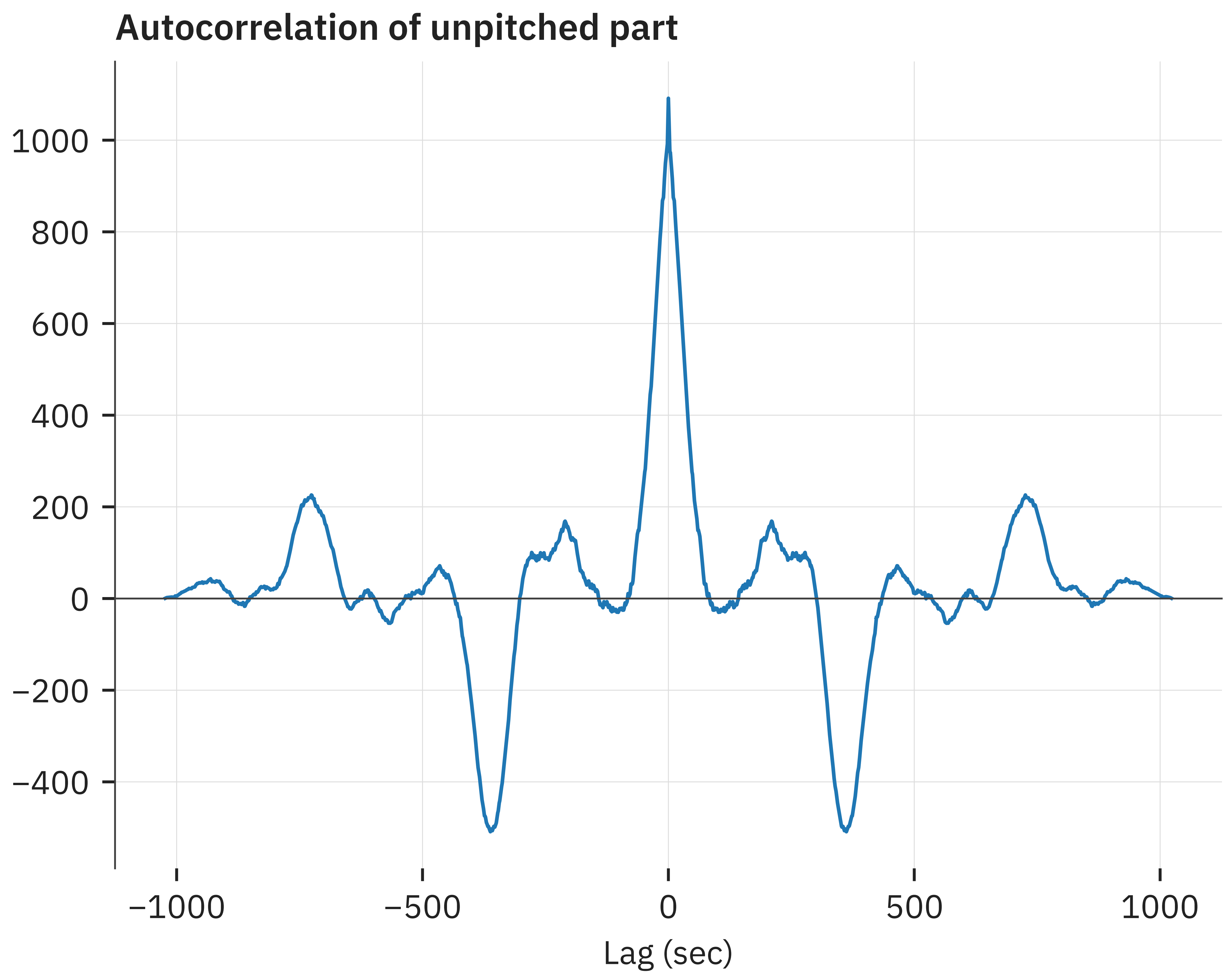
- Not all areas are pitched
  - We need a pitched/unpitched detector



Using the autocorrelation peak

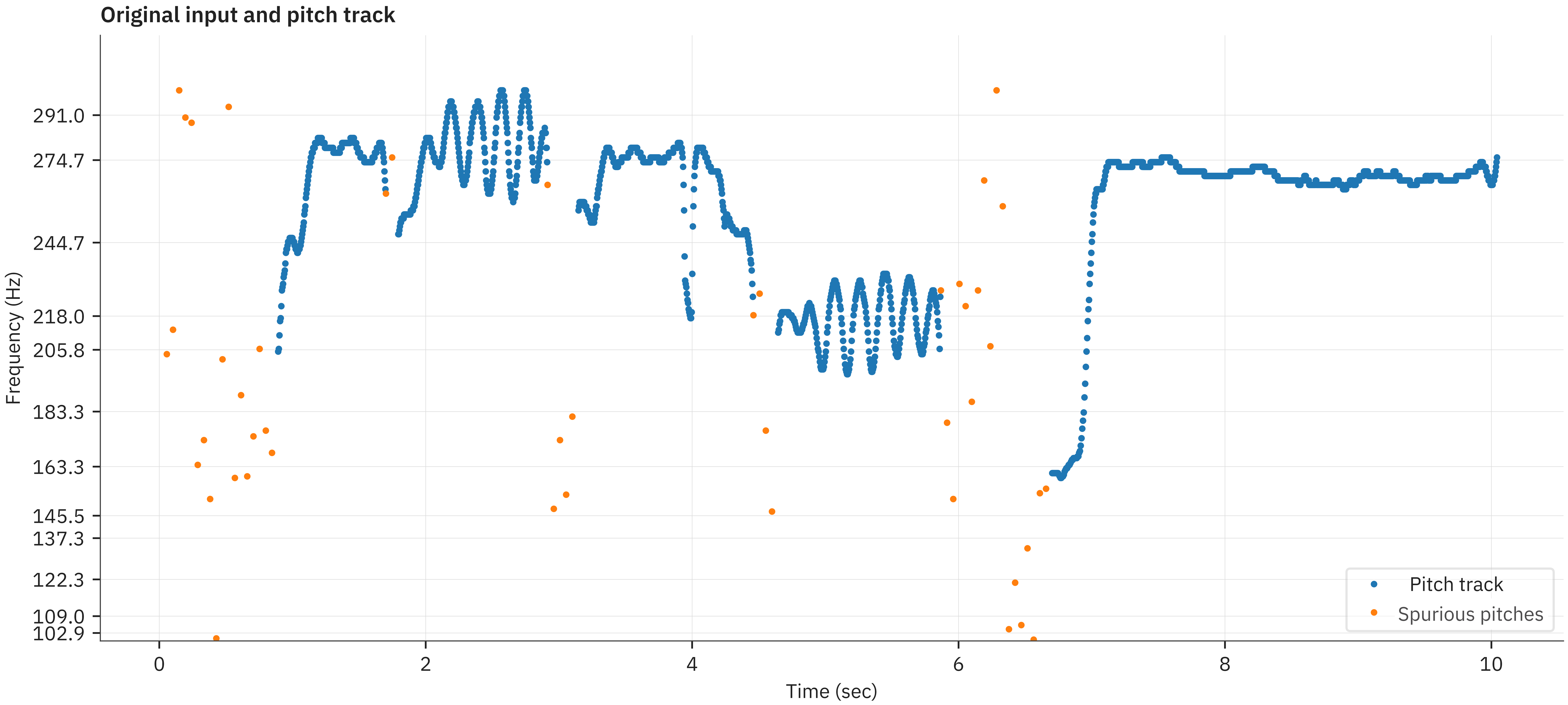
- Height of first peak is an indication of similarity across periods, i.e. "pitchness"





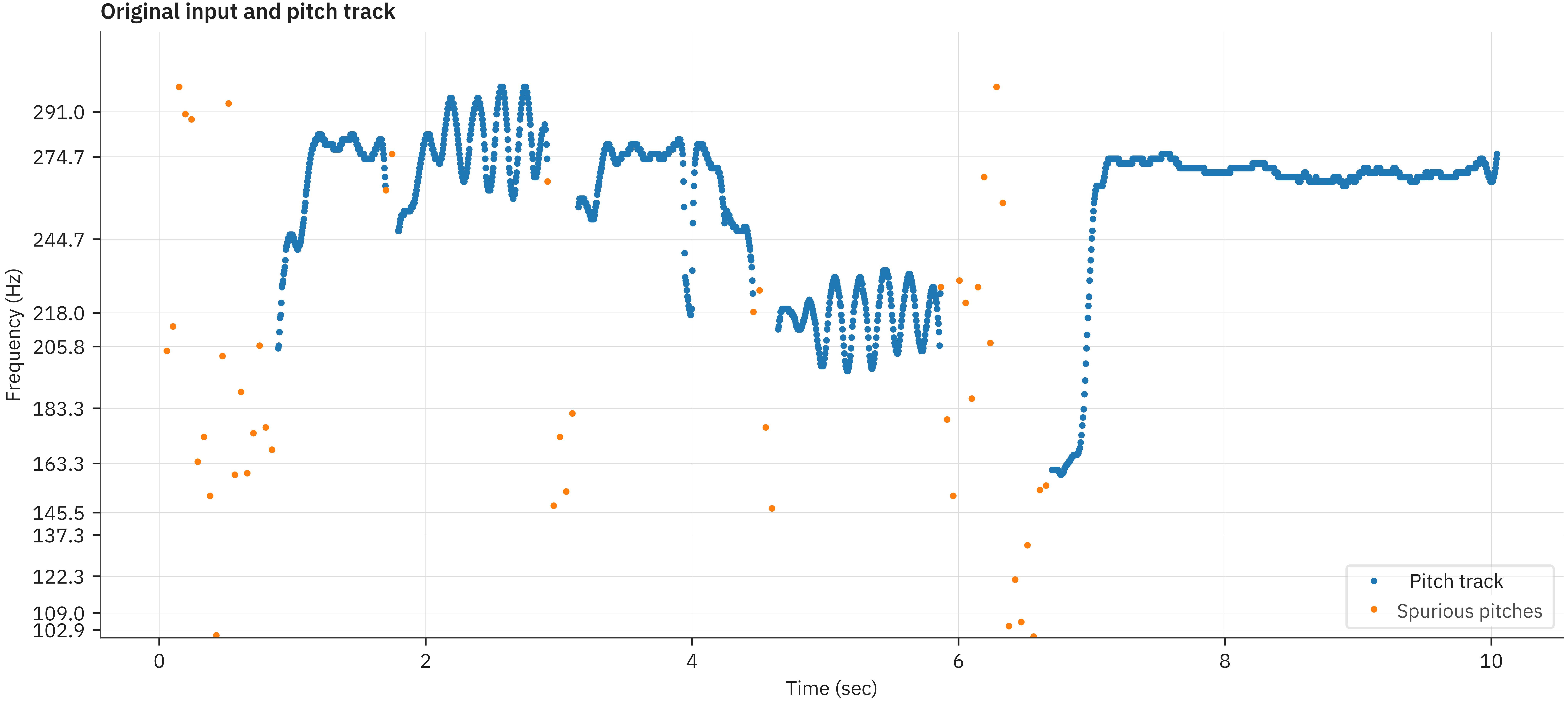
Pitch track, with pitched decisions

- Blue points indicate strong pitch
  - Orange points indicate weak pitch



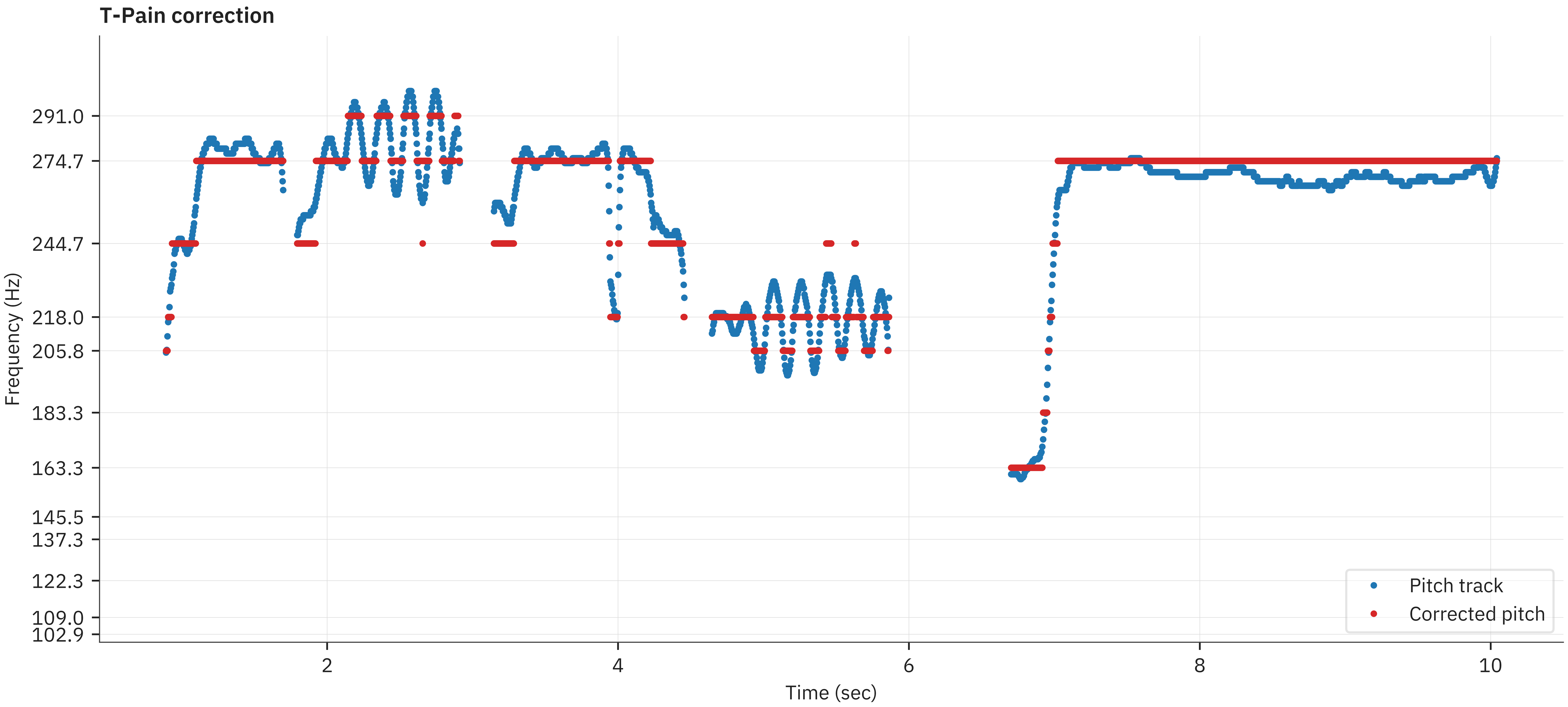
Correcting the input

- We want the pitch to lie on the musical scale axis
  - Preselected frequencies which are "right"



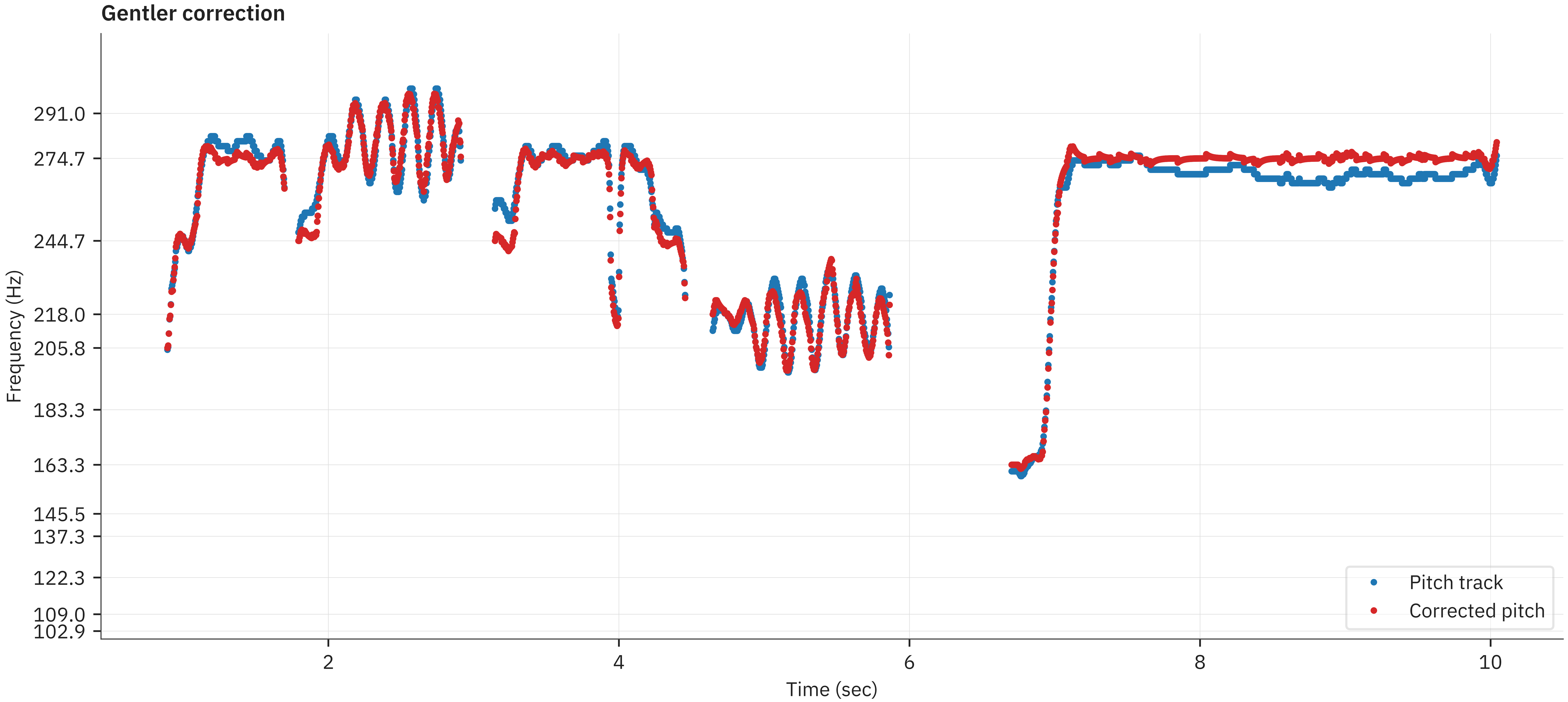
Using PSOLA we can correct pitch

- But we need to be gentle
  - Otherwise we get abrupt jumps (the Cher/T-Pain effect)



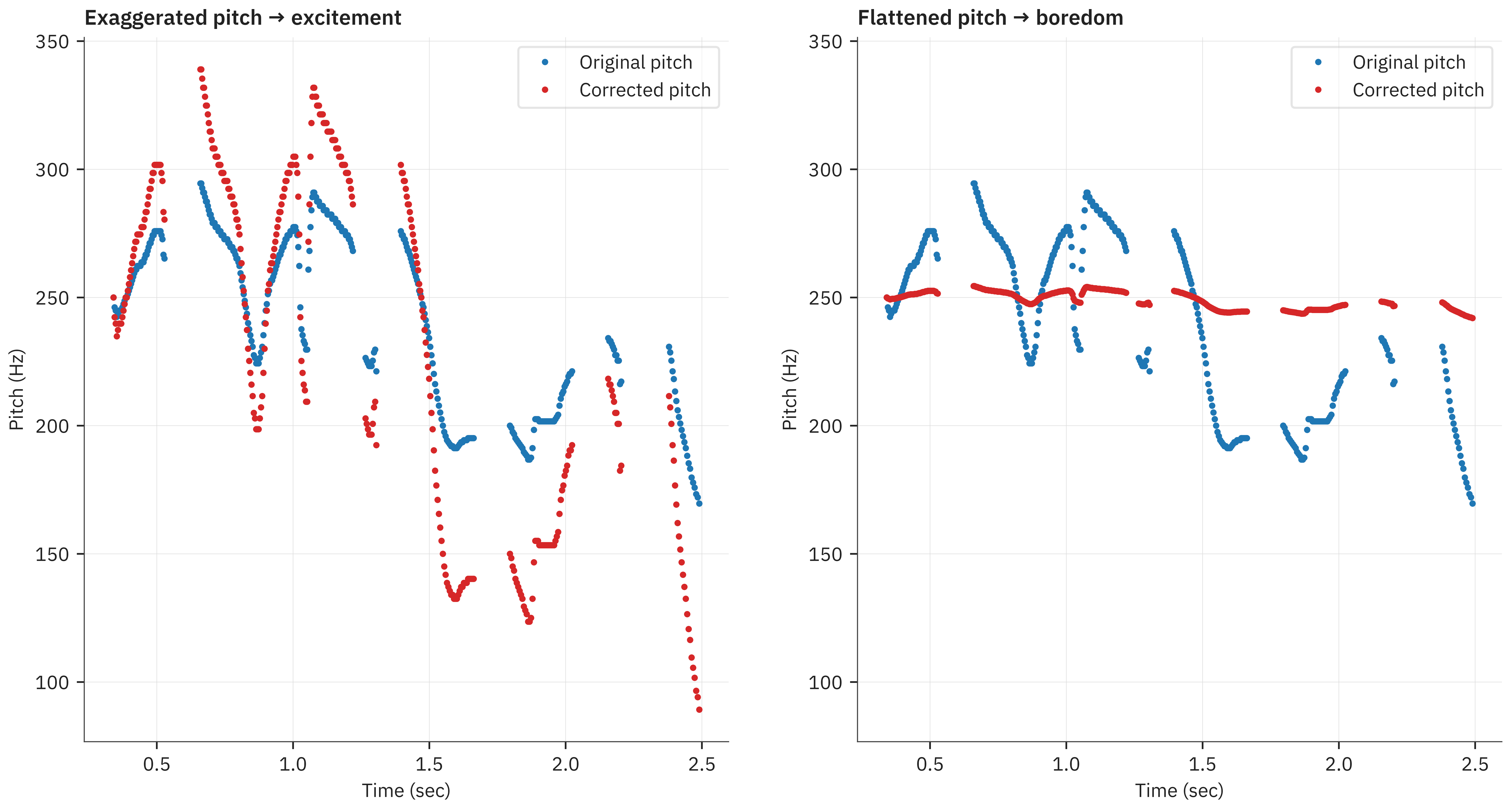
Gentler correction

- Instead, nudge the pitches towards the right value
  - But don't get there exactly



Changing spoken pitch intonation

- We can also use this for “emotion changing”
  - We can exaggerate pitch to create excitement
  - We can flatten pitch to create boredom



Recap

- Pitch and its definition
- Pitch estimation methods
  - Simple: zero/peak counting
  - Fancier: autocorrelation/cepstrum
- Pitch modifications
  - PSOLA
  - Designing an autotuner

Reading material

- Pitch extraction and fundamental frequency: History and current techniques
  - https://www.cs.bu.edu/fac/snyder/cs583/Literature and Resources/PitchExtractionMastersThesis.pdf
- The YIN pitch estimator
  - http://audition.ens.fr/adc/pdf/2002_JASA_YIN.pdf
- Voice transformation via PSOLA
  - https://ieeexplore.ieee.org/document/225951